LAPORAN

PROJEK AKHIR PRAKTIKUM

ALGORITMA PEMROGRAMAN LANJUT

TOKO SKINCARE

KELAS IF C1’25

KELOMPOK 2

DAPA SI IMUP

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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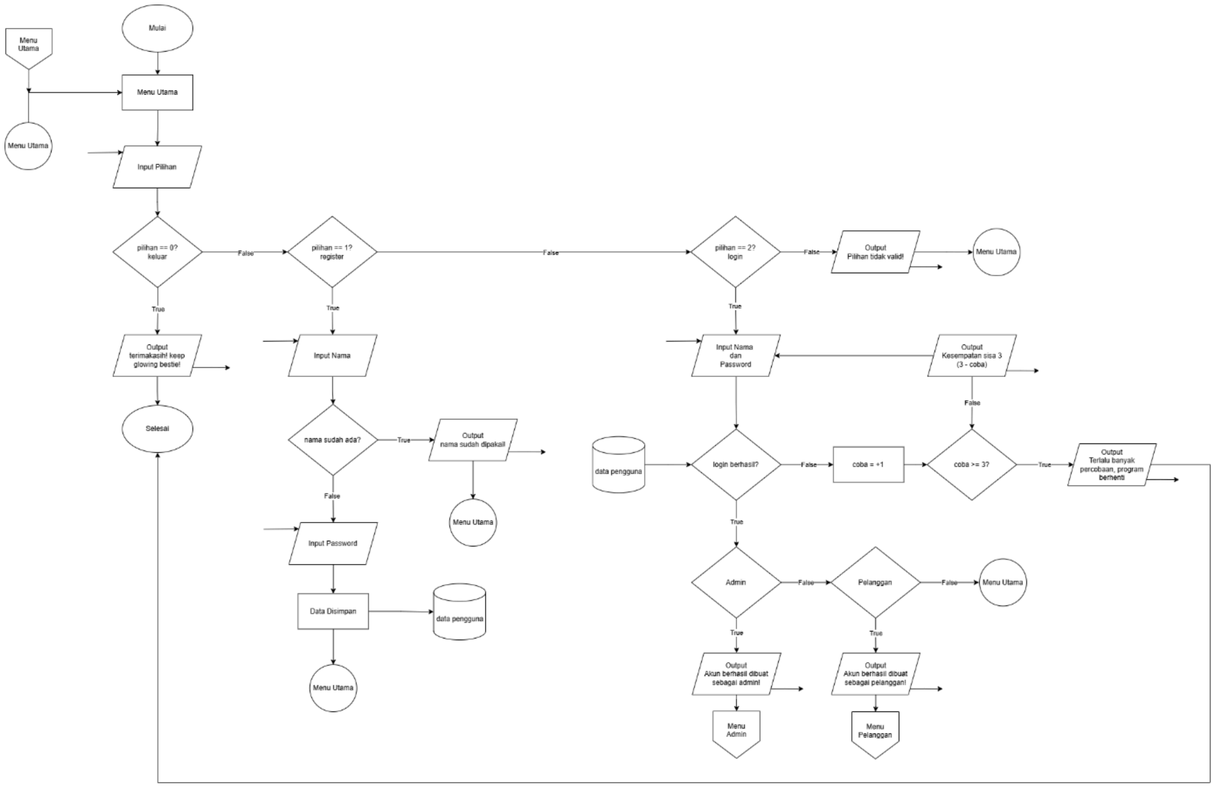
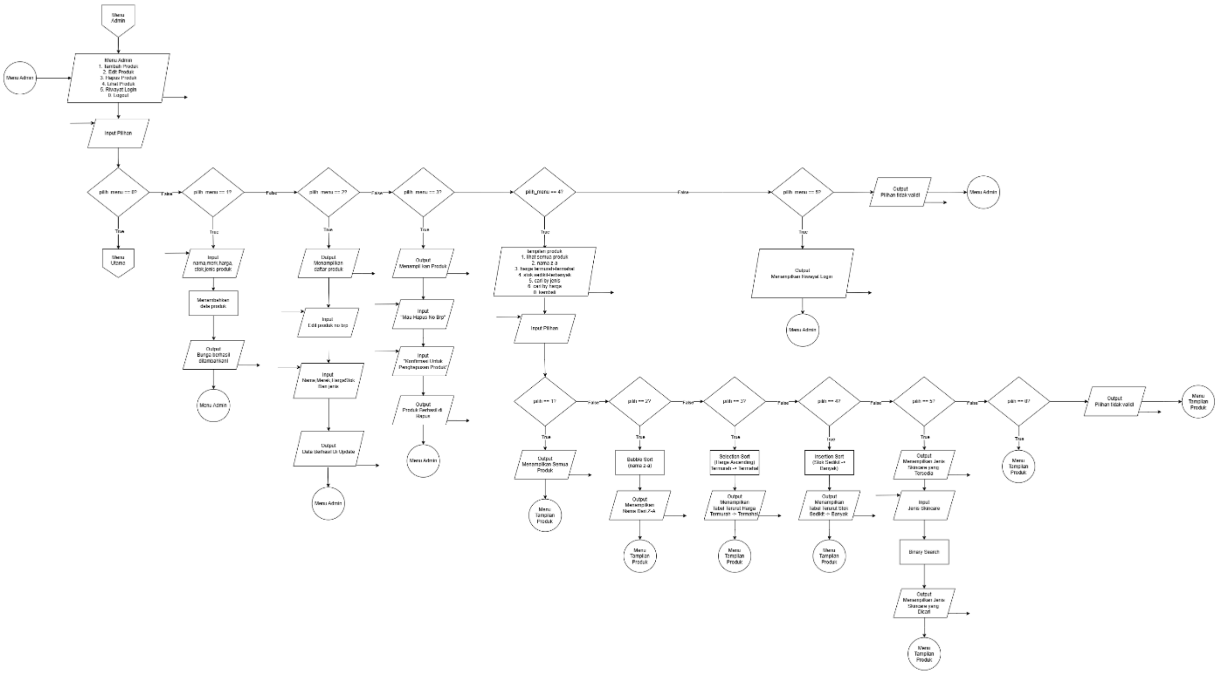
1. Flowchart

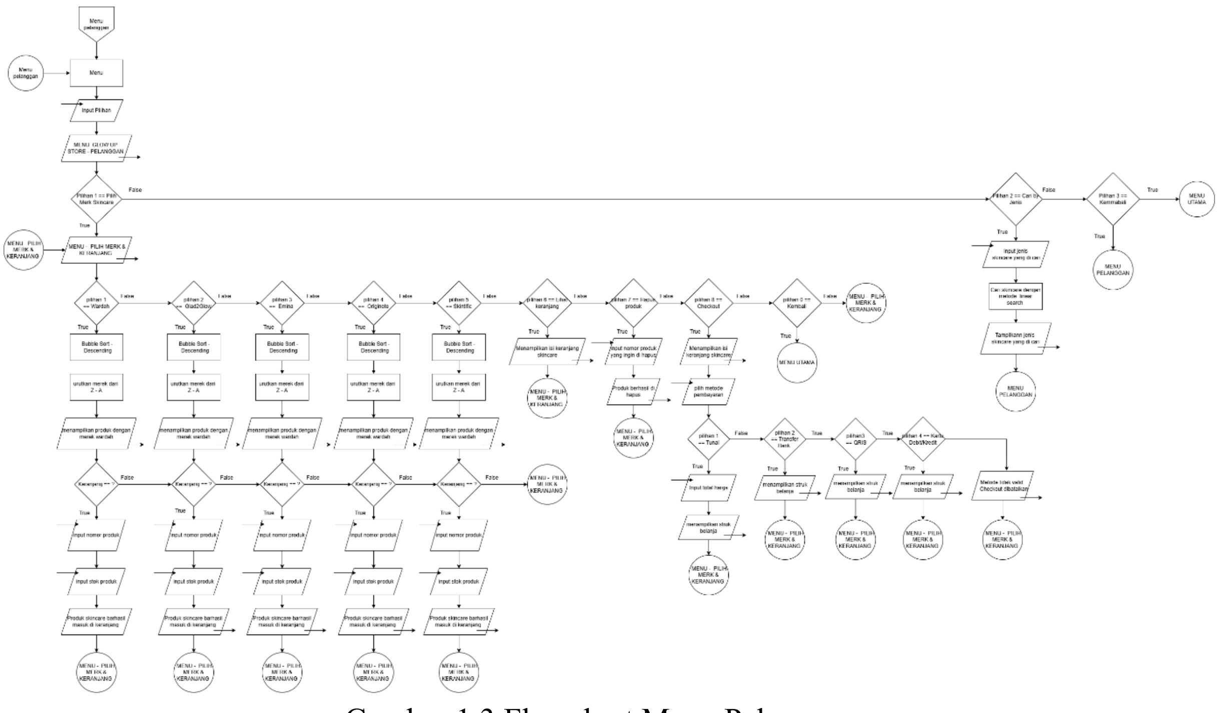
- Menu Utama : Pengguna melakukan login, lalu sistem membaca data akun di file teks untuk menentukan hak akses (Admin atau Pelanggan).
- Admin : Masuk ke menu pengelolaan, Proses tambah, edit, hapus, lihat produk dilakukan. Data produk di toko otomatis diperbarui.
- Pelanggan : Masuk ke menu belanja, Proses pilih produk, masuk keranjang, dan melakukan pembayaran. Riwayat transaksi otomatis tersimpan ke file teks.



Gambar 1.1 Flowchart Menu Utama



Gambar 1.2 Flowchart Menu Admin



Gambar 1.3 Flowchart Menu Pelanggan

2. Deskripsi Singkat Program

Program ini bisa digunakan oleh dua jenis pengguna, admin yang ngurusin data bunganya, sama pelanggan yang mau beli atau cari bunga. Datanya menyimpan di memori selama program berjalan, jadi kalau programnya di tutup maka data tersebut hilang.

3. Souce Code

```
#include <iostream>
#include <iomanip>
#include <ctime>
#include <fstream>
#include <sstream>
#include <stdexcept>
#include <cctype>

using namespace std;

// ANSI COLOR CODES
#define RESET       "\033[0m"
#define BOLD        "\033[1m"
#define BRED        "\033[91m"
#define BGREEN      "\033[92m"
#define BYELLOW     "\033[93m"
#define BBLUE       "\033[94m"
#define BMAGENTA    "\033[95m"
#define BCYAN       "\033[96m"
#define BWHITE      "\033[97m"
#define BG_MAGENTA  "\033[45m"
#define BG_CYAN     "\033[46m"
#define BG_BLUE     "\033[44m"
#define BG_BLACK    "\033[40m"

// WARNA HARGA
#define BHARGA      "\033[93m"   // BYELLOW — dipakai di SEMUA tampilan
harga

// LEBAR KOLOM PRODUK
#define TBL_SEP
"+-----+-------------------------+-------------+-----------------+-------+--
-------------+"
#define TBL_HDR "| %-3s | %-23s | %-11s | %-15s | %-5s | %-13s |\n"
#define TBL_ROW "| %s%-3d%s | %s%-23s%s | %s%-11s%s | %s%-15s%s | %s%-5d%s |
%s%-13s%s |\n"

// TAMPILAN
void cetakGaris(string warna, int panjang = 56) {
    cout << warna;
    for (int i = 0; i < panjang; i++) cout << "=";
    cout << RESET << "\n";
}
```

```
void cetakGarisTipis(string warna, int panjang = 56) {
    cout << warna;
    for (int i = 0; i < panjang; i++) cout << "-";
    cout << RESET << "\n";
}

void cetakBaris(string warna, string teks, int panjang = 56) {
    int isi = panjang - 2;
    cout << warna << "| " << RESET << left << setw(isi - 2) << teks << warna
<< " |" << RESET << "\n";
}

void cetakTengah(string bgWarna, string teksWarna, string teks, int panjang
= 56) {
    int isi = panjang - 2;
    int pad = (isi - (int)teks.size()) / 2;
    if (pad < 0) pad = 0;
    int padKanan = isi - (int)teks.size() - pad;
    if (padKanan < 0) padKanan = 0;
    cout << bgWarna << "|";
    for (int i = 0; i < pad; i++) cout << " ";
    cout << teksWarna << BOLD << teks << RESET << bgWarna;
    for (int i = 0; i < padKanan; i++) cout << " ";
    cout << "|" << RESET << "\n";
}

void cetakHeader(string judul, string warna, int panjang = 56) {
    cetakGaris(warna, panjang);
    cetakTengah(warna, BWHITE, judul, panjang);
    cetakGaris(warna, panjang);
}

// WAKTU & TRANSAKSI
string getWaktuSekarang() {
    time_t now = time(0);
    struct tm *ltm = localtime(&now);
    char buf[80];
    sprintf(buf, "%02d/%02d/%04d %02d:%02d:%02d",
        ltm->tm_mday, ltm->tm_mon + 1, ltm->tm_year + 1900,
        ltm->tm_hour, ltm->tm_min, ltm->tm_sec);
    return string(buf);
}

string generateNoTransaksi() {
```

```
    time_t now = time(0);
    struct tm *ltm = localtime(&now);
    char buf[40];
    sprintf(buf, "TRX%04d%02d%02d%02d%02d%02d",
        ltm->tm_year + 1900, ltm->tm_mon + 1, ltm->tm_mday,
        ltm->tm_hour, ltm->tm_min, ltm->tm_sec);
    return string(buf);
}

const string FILE_PENGGUNA      = "data_pengguna.txt";
const string FILE_RIWAYAT_LOGIN = "riwayat_login.txt";

struct Harga { double satuan; string matauang; };
struct Stok  { int jumlah;   string satuan;   };
struct Produk {
    string nama, merk;
    Harga  harga;
    Stok   stok;
    string jenis;
    int    aktif;
};
struct Pengguna {
    string nama, password, role;
    int    aktif;
};
struct ItemKeranjang {
    int idxProduk;
    int jumlah;
};

int MAKS_PENGGUNA  = 10;
int MAKS_PRODUK    = 100;
int MAKS_KERANJANG = 50;

Pengguna dataPengguna[10];
Produk   daftarProduk[100];
int jumlahPengguna = 0;
int totalProduk    = 0;

// SIMPAN & LOAD PENGGUNA
void simpanPengguna() {
    try {
        ofstream f(FILE_PENGGUNA.c_str());
        if (!f.is_open()) throw runtime_error("Gagal membuka file
```

```
pengguna.");
        f << jumlahPengguna << "\n";
        for (int i = 0; i < jumlahPengguna; i++)
            f << dataPengguna[i].nama     << "\n"
              << dataPengguna[i].password << "\n"
              << dataPengguna[i].role     << "\n"
              << dataPengguna[i].aktif    << "\n";
        f.close();
    } catch (const runtime_error& e) {
        cout << BRED << "  [!] Error: " << e.what() << RESET << "\n";
    }
}

void loadPengguna() {
    try {
        ifstream f(FILE_PENGGUNA.c_str());
        if (!f.is_open()) return;
        int jml = 0;
        f >> jml; f.ignore();
        for (int i = 0; i < jml && jumlahPengguna < MAKS_PENGGUNA; i++) {
            string nama, password, role; int aktif = 1;
            getline(f, nama); getline(f, password); getline(f, role);
            f >> aktif; f.ignore();
            if (nama.empty() || password.empty() || role.empty()) continue;
            int dup = 0;
            for (int j = 0; j < jumlahPengguna; j++)
                if (dataPengguna[j].nama == nama) { dup = 1; break; }
            if (dup) continue;
            dataPengguna[jumlahPengguna++] = {nama, password, role, aktif};
        }
        f.close();
    } catch (const exception& e) {
        cout << BRED << "  [!] Gagal load pengguna: " << e.what() << RESET
<< "\n";
    }
}

// RIWAYAT LOGIN
void simpanRiwayatLogin(string namaUser, string role, string status) {
    try {
        ofstream f(FILE_RIWAYAT_LOGIN.c_str(), ios::app);
        if (!f.is_open()) throw runtime_error("Gagal membuka file riwayat
login.");
        f << getWaktuSekarang() << "|" << namaUser << "|" << role << "|" <<
```

```
status << "\n";
        f.close();
    } catch (const runtime_error& e) {
        cout << BRED << "  [!] Error: " << e.what() << RESET << "\n";
    }
}

// TABEL RIWAYAT LOGIN
#define LOG_SEP
"+----------------------+------------------+-------------+----------+"
#define LOG_HDR "| %-20s | %-17s | %-12s | %-9s|\n"
#define LOG_ROW "| %-20s | %-17s | %s%-12s%s | %s%-9s%s|\n"

void tampilRiwayatLogin() {
    try {
        ifstream f(FILE_RIWAYAT_LOGIN.c_str());
        if (!f.is_open()) throw runtime_error("File riwayat login tidak
ditemukan.");

        cout << BCYAN << LOG_SEP << "\n" << RESET;
        printf(LOG_HDR,
            (BYELLOW + string("Waktu") + RESET).c_str(),
            (BGREEN  + string("Nama") + RESET).c_str(),
            (BMAGENTA+ string("Role") + RESET).c_str(),
            (BWHITE  + string("Status") + RESET).c_str());
        cout << BCYAN << LOG_SEP << "\n" << RESET;

        string baris; int ada = 0;
        while (getline(f, baris)) {
            string kolom[4]; int k = 0; string tmp = "";
            for (int i = 0; i < (int)baris.size(); i++) {
                if (baris[i] == '|' && k < 3) { kolom[k++] = tmp; tmp = "";
}
                else tmp += baris[i];
            }
            kolom[k] = tmp;
            if ((int)kolom[1].size() > 17) kolom[1] = kolom[1].substr(0, 14)
+ "...";
            if ((int)kolom[2].size() > 12) kolom[2] = kolom[2].substr(0, 9)
+ "...";
            string sc = (kolom[3].find("BERHASIL") != string::npos) ? BGREEN
: BRED;
            cout << BCYAN << "| " << RESET
                 << left << setw(20) << kolom[0]
```

```
                << BCYAN << " | " << RESET
                << setw(17) << kolom[1]
                << BCYAN << " | " << BMAGENTA
                << setw(12) << kolom[2]
                << BCYAN << " | " << sc
                << setw(9) << kolom[3]
                << BCYAN << "|\n" << RESET;
            ada = 1;
        }
        if (!ada) {
            cout << BCYAN << "| " << BYELLOW
                << left << setw(58) << "(belum ada riwayat login)"
                << BCYAN << "|\n" << RESET;
        }
        cout << BCYAN << LOG_SEP << "\n" << RESET;
        f.close();
    } catch (const runtime_error& e) {
        cout << BRED << "  [!] " << e.what() << RESET << "\n";
    }
}

void menuAdmin(string namaUser);
void menuPelanggan(string namaUser);
int  doLogin(Pengguna* data, int jml);

// HELPER ATAU TOOLS
string angkaKeStr(long long n) {
    if (n == 0) return "0";
    string s = "";
    while (n > 0) { s = char('0' + n % 10) + s; n /= 10; }
    return s;
}

string rupiahFormat(double angka) {
    string s = angkaKeStr((long long)angka);
    int n = s.size(); string hasil = "";
    for (int i = 0; i < n; i++) {
        if (i > 0 && (n - i) % 3 == 0) hasil += ".";
        hasil += s[i];
    }
    return "Rp " + hasil;
}

int validAngka(string s) {
```

```
    if (s.empty()) return 0;
    for (int i = 0; i < (int)s.size(); i++)
        if (s[i] < '0' || s[i] > '9') return 0;
    return 1;
}

// VALIDASI BARU
// Cek apakah string mengandung angka
bool containsDigit(const string& s) {
    for (char c : s) {
        if (isdigit(c)) return true;
    }
    return false;
}

// Cek apakah string mengandung huruf
bool containsLetter(const string& s) {
    for (char c : s) {
        if (isalpha(c)) return true;
    }
    return false;
}

int bacaAngka(string s) {
    if (s.empty()) throw invalid_argument("Input tidak boleh kosong.");
    if (!validAngka(s)) throw invalid_argument("Input harus berupa angka
positif (tanpa huruf).");
    int hasil = 0;
    for (int i = 0; i < (int)s.size(); i++) hasil = hasil * 10 + (s[i] -
'0');
    return hasil;
}

double bacaHarga(string s) {
    if (s.empty()) throw invalid_argument("Harga tidak boleh kosong.");
    if (!validAngka(s)) throw invalid_argument("Harga harus berupa angka
positif (tanpa huruf).");
    double h = 0;
    for (int i = 0; i < (int)s.size(); i++) h = h * 10 + (s[i] - '0');
    if (h <= 0) throw out_of_range("Harga harus lebih dari 0.");
    return h;
}

void cetakHargaKolom(string harga, int w) {
```

```
    int pad = w - (int)harga.size();
    if (pad < 0) pad = 0;
    cout << BHARGA;
    for (int i = 0; i < pad; i++) cout << " ";
    cout << harga << RESET;
}

// TABEL PRODUK
void cetakHeaderTabel() {
    cout << BCYAN <<
"+-----+-------------------------+-------------+-----------------+-------+--
-------------+\n";
    cout << "| " << BYELLOW   << left  << setw(3)  << "No"
         << BCYAN << " | " << BGREEN   << left  << setw(23) << "Nama Produk"
         << BCYAN << " | " << BMAGENTA << left  << setw(11) << "Merk"
         << BCYAN << " | " << BHARGA   << right << setw(15) << "Harga"
         << BCYAN << " | " << BWHITE   << left  << setw(5)  << "Stok"
         << BCYAN << " | " << BBLUE    << left  << setw(13) << "Jenis"
         << BCYAN << " |\n";
    cout <<
"+-----+-------------------------+-------------+-----------------+-------+--
-------------+\n" << RESET;
}

void cetakBarisProdukTabel(int no, Produk &p) {
    string nama = p.nama;
    if ((int)nama.size() > 23) nama = nama.substr(0, 20) + "...";
    string merk = p.merk;
    if ((int)merk.size() > 11) merk = merk.substr(0, 8) + "...";
    string jenis = p.jenis;
    if ((int)jenis.size() > 13) jenis = jenis.substr(0, 10) + "...";
    string harga = rupiahFormat(p.harga.satuan);
    if ((int)harga.size() > 15) harga = harga.substr(0, 12) + "...";

string sc = (p.stok.jumlah <= 10) ? BRED : BGREEN;

    cout << BCYAN << "| " << BYELLOW   << left  << setw(3)  << no
         << BCYAN << " | " << BWHITE   << left  << setw(23) << nama
         << BCYAN << " | " << BMAGENTA << left  << setw(11) << merk
         << BCYAN << " | ";
    cetakHargaKolom(harga, 15);
    cout << BCYAN << " | " << sc       << left  << setw(5)  << p.stok.jumlah
         << BCYAN << " | " << BBLUE    << left  << setw(13) << jenis
         << BCYAN << " |\n" << RESET;
```

```
}

void cetakFooterTabel() {
    cout << BCYAN <<
"+-----+-------------------------+-------------+-----------------+-------+--
-------------+\n" << RESET;
}

int tampilTabel(int stockOnly) {
    cetakHeaderTabel();
    int no = 0, adaData = 0;
    for (int i = 0; i < totalProduk; i++) {
        if (daftarProduk[i].aktif == 0) continue;
        if (stockOnly == 1 && daftarProduk[i].stok.jumlah == 0) continue;
        no++;
        cetakBarisProdukTabel(no, daftarProduk[i]);
        adaData = 1;
    }
    if (!adaData) {
        cout << BCYAN << "|   " << BYELLOW << left << setw(86) << "(belum ada
data produk)"
             << BCYAN << "|\n" << RESET;
    }
    cetakFooterTabel();
    return no;
}

// SORTING

void selectionSortHargaAsc(Produk arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int idxMin = i;
        for (int j = i + 1; j < n; j++)
            if (arr[j].harga.satuan < arr[idxMin].harga.satuan) idxMin = j;
        if (idxMin != i) { Produk t = arr[i]; arr[i] = arr[idxMin];
arr[idxMin] = t; }
    }
}
void bubbleSortNamaDesc(Produk arr[], int n) {
    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - i - 1; j++)
            if (arr[j].nama < arr[j+1].nama) { Produk t = arr[j]; arr[j] =
arr[j+1]; arr[j+1] = t; }
}
```

```
void insertionSortStokAsc(Produk arr[], int n) {
    for (int i = 1; i < n; i++) {
        Produk k = arr[i]; int j = i - 1;
        while (j >= 0 && arr[j].stok.jumlah > k.stok.jumlah) { arr[j+1] =
arr[j]; j--; }
        arr[j+1] = k;
    }
}

void tampilDenganSorting(int mode) {
    Produk tmp[100]; int jml = 0;
    for (int i = 0; i < totalProduk; i++)
        if (daftarProduk[i].aktif) tmp[jml++] = daftarProduk[i];
    if (jml == 0) { cout << BYELLOW << "  Belum ada data produk.\n" <<
RESET; return; }
    if      (mode == 1) bubbleSortNamaDesc(tmp, jml);
    else if (mode == 2) selectionSortHargaAsc(tmp, jml);
    else if (mode == 3) insertionSortStokAsc(tmp, jml);
    cetakHeaderTabel();
    for (int i = 0; i < jml; i++) cetakBarisProdukTabel(i + 1, tmp[i]);
    cetakFooterTabel();
}

int pilihNomor(string aksi) {
    int peta[100], no = 0;
    for (int i = 0; i < totalProduk; i++)
        if (daftarProduk[i].aktif) peta[no++] = i;
    tampilTabel(0);
    if (no == 0) { cout << BYELLOW << "  Belum ada produk.\n" << RESET;
return -1; }
    try {
        string pilih;
        cout << BGREEN << "\n  " << aksi << " produk nomor berapa? " <<
BYELLOW;
        getline(cin, pilih); cout << RESET;
        int nomor = bacaAngka(pilih);
        if (nomor < 1 || nomor > no)
            throw out_of_range("Pilih antara 1 sampai " + angkaKeStr(no) +
".");
        return peta[nomor - 1];
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const out_of_range& e)       { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
```

```cpp
    return -1;
}

// FITUR ADMIN

void tambahProduk() {
    cetakHeader("TAMBAH PRODUK BARU", BGREEN);
    if (totalProduk >= MAKS_PRODUK) { cout << BRED << "  [!] Data produk sudah penuh!\n" << RESET; return; }
    try {
        Produk p; string hargaStr, stokStr;

// Input Nama - tidak boleh ada angka
        cout << BCYAN << "  Nama produk : " << BWHITE;
        getline(cin, p.nama);
        if (p.nama.empty()) throw invalid_argument("Nama produk tidak boleh kosong.");
        if (containsDigit(p.nama)) throw invalid_argument("Nama produk tidak boleh mengandung angka!");

// Input Merk - tidak boleh ada angka
        cout << BCYAN << "  Merk : " << BWHITE;
        getline(cin, p.merk);
        if (p.merk.empty()) throw invalid_argument("Merk tidak boleh kosong.");
        if (containsDigit(p.merk)) throw invalid_argument("Merk tidak boleh mengandung angka!");

// Input Harga - harus angka
        cout << BCYAN << "  Harga (Rp) : " << BHARGA;
        getline(cin, hargaStr);
        if (hargaStr.empty()) throw invalid_argument("Harga tidak boleh kosong.");
        if (containsLetter(hargaStr)) throw invalid_argument("Harga harus berupa angka, tidak boleh ada huruf!");

// Input Stok - harus angka
        cout << BCYAN << "  Stok : " << BWHITE;
        getline(cin, stokStr);
        if (stokStr.empty()) throw invalid_argument("Stok tidak boleh kosong.");
        if (containsLetter(stokStr)) throw invalid_argument("Stok harus berupa angka, tidak boleh ada huruf!");
```

```
// Input Jenis - tidak boleh ada angka
        cout << BCYAN << "  Jenis       : " << BBLUE;
        getline(cin, p.jenis);
        cout << RESET;
        if (p.jenis.empty()) throw invalid_argument("Jenis tidak boleh
kosong.");
        if (containsDigit(p.jenis)) throw invalid_argument("Jenis tidak
boleh mengandung angka!");

// Konversi dan simpan
        p.harga.satuan = bacaHarga(hargaStr);
        p.harga.matauang = "IDR";
        p.stok.jumlah = bacaAngka(stokStr);
        p.stok.satuan = "pcs";
        p.aktif = 1;
        daftarProduk[totalProduk++] = p;
        cout << BGREEN << "  [OK] Produk '" << p.nama << "' berhasil
ditambahkan!\n" << RESET;
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const out_of_range& e)       { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
}

void editProduk() {
    cetakHeader("EDIT DATA PRODUK", BYELLOW);
    int idx = pilihNomor("Edit");
    if (idx == -1) return;
    try {
        Produk &p = daftarProduk[idx];
        cout << BYELLOW << "\n  Edit produk (Enter = tidak berubah)\n" <<
RESET;
        string nb, mb, hb, sb, jb;

        cout << BCYAN << "  Nama  [" << BWHITE   << p.nama << BCYAN << "]: "
<< BWHITE;
        getline(cin, nb);
        if (!nb.empty()) {
            if (containsDigit(nb)) throw invalid_argument("Nama tidak boleh
mengandung angka!");
            p.nama = nb;
        }

        cout << BCYAN << "  Merk  [" << BMAGENTA << p.merk << BCYAN << "]: "
```

```
<< BWHITE;
        getline(cin, mb);
        if (!mb.empty()) {
            if (containsDigit(mb)) throw invalid_argument("Merk tidak boleh
mengandung angka!");
            p.merk = mb;
        }

        cout << BCYAN << "  Harga [" << BHARGA   <<
rupiahFormat(p.harga.satuan) << BCYAN << "]: " << BHARGA;
        getline(cin, hb);
        if (!hb.empty()) {
            if (containsLetter(hb)) throw invalid_argument("Harga harus
berupa angka!");
            try { p.harga.satuan = bacaHarga(hb); }
            catch(const exception& e) { cout << BRED << "  [!] Harga tidak
diubah.\n" << RESET; }
        }

        cout << BCYAN << "  Stok  [" << BGREEN   << p.stok.jumlah << BCYAN
<< "]: " << BWHITE;
        getline(cin, sb);
        if (!sb.empty()) {
            if (containsLetter(sb)) throw invalid_argument("Stok harus
berupa angka!");
            try { p.stok.jumlah = bacaAngka(sb); }
            catch(const exception& e) { cout << BRED << "  [!] Stok tidak
diubah.\n" << RESET; }
        }

        cout << BCYAN << "  Jenis [" << BBLUE    << p.jenis << BCYAN << "]:
" << BWHITE;
        getline(cin, jb);
        cout << RESET;
        if (!jb.empty()) {
            if (containsDigit(jb)) throw invalid_argument("Jenis tidak boleh
mengandung angka!");
            p.jenis = jb;
        }

        cout << BGREEN << "  [OK] Data berhasil diupdate!\n" << RESET;
    } catch (const exception& e) { cout << BRED << "  [!] " << e.what() <<
RESET << "\n"; }
}
```

```
void nonaktifkanProduk(int *s, string nama) {
    if (!s) throw runtime_error("Pointer null!");
    *s = 0;
    cout << BGREEN << "  [OK] Produk '" << nama << "' dihapus!\n" << RESET;
}

void hapusProduk() {
    cetakHeader("HAPUS PRODUK", BRED);
    int idx = pilihNomor("Hapus");
    if (idx == -1) return;
    try {
        string k;
        while (true) {
            cout << BYELLOW << "  Yakin hapus '" << BWHITE <<
daftarProduk[idx].nama << BYELLOW << "'? (y/n): " << BWHITE;
            getline(cin, k); cout << RESET;
            if (k=="y"||k=="Y"||k=="n"||k=="N") break;
            cout << BRED << "  Input tidak valid!\n" << RESET;
        }
        if (k=="y"||k=="Y") nonaktifkanProduk(&daftarProduk[idx].aktif,
daftarProduk[idx].nama);
        else cout << BYELLOW << "  Dibatalkan.\n" << RESET;
    } catch (const runtime_error& e) { cout << BRED << "  [!] " << e.what()
<< RESET << "\n"; }
}

// FITUR PELANGGAN
void cariNama() {
    cetakHeader("CARI PRODUK BY JENIS", BCYAN);
    int jml = 0;
    for (int i = 0; i < totalProduk; i++) if (daftarProduk[i].aktif) jml++;
    if (jml == 0) { cout << BYELLOW << "  Belum ada produk.\n" << RESET;
return; }
    cout << BCYAN << "  Jenis tersedia : " << RESET;
    cout << BMAGENTA << "Serum" << RESET << ", " << BMAGENTA << "Toner" <<
RESET << ", "
         << BMAGENTA << "Moisturizer" << RESET << ", " << BMAGENTA << "Face
Wash" << RESET << ", "
         << BMAGENTA << "Sunscreen" << RESET << "\n";
    cout << BCYAN << "  (ketik " << BYELLOW << "semua" << BCYAN << " untuk
tampilkan semua jenis)\n" << RESET;
    cout << "\n";
    try {
```

```
        string keyword;
        cout << BGREEN << "  Cari jenis skincare: " << BYELLOW;
        getline(cin, keyword); cout << RESET;
        if (keyword.empty()) throw invalid_argument("Input tidak boleh
kosong.");
        string kwLower = keyword;
        for (int i = 0; i < (int)kwLower.size(); i++) kwLower[i] =
tolower(kwLower[i]);
        bool cariSemua = (kwLower == "semua");
        cetakHeaderTabel();
        int ketemu = 0;
        for (int i = 0; i < totalProduk; i++) {
            if (!daftarProduk[i].aktif) continue;
            string jpLower = daftarProduk[i].jenis;
            for (int j = 0; j < (int)jpLower.size(); j++) jpLower[j] =
tolower(jpLower[j]);
            bool cocok = cariSemua || (jpLower.find(kwLower) !=
string::npos);
            if (!cocok) continue;
            ketemu++;
            cetakBarisProdukTabel(ketemu, daftarProduk[i]);
        }
        if (ketemu == 0) {
            cout << BCYAN << "|  " << BRED << left << setw(86)
                 << ("(tidak ada produk untuk jenis \"" + keyword + "\")")
                 << BCYAN << "|\n" << RESET;
            cetakFooterTabel();
            cout << BYELLOW << "\n  Mungkin maksud kamu salah satu ini:\n"
<< RESET;
            string jenisUnik[20]; int jmlJenis = 0;
            for (int i = 0; i < totalProduk; i++) {
                if (!daftarProduk[i].aktif) continue;
                bool sudahAda = false;
                for (int j = 0; j < jmlJenis; j++)
                    if (jenisUnik[j] == daftarProduk[i].jenis) { sudahAda =
true; break; }
                if (!sudahAda) jenisUnik[jmlJenis++] =
daftarProduk[i].jenis;
            }
            for (int i = 0; i < jmlJenis; i++)
                cout << BMAGENTA << "    -> " << BWHITE << jenisUnik[i] <<
"\n" << RESET;
        } else {
            cetakFooterTabel();
```

```
            cout << BGREEN << "  [OK] Ditemukan " << ketemu << " produk
untuk jenis \""
                 << keyword << "\".\n" << RESET;
        }
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
}

void cariHarga() {
    cetakHeader("CARI PRODUK BY HARGA", BYELLOW);
    Produk tmp[100]; int jml = 0;
    for (int i = 0; i < totalProduk; i++) if (daftarProduk[i].aktif)
tmp[jml++] = daftarProduk[i];
    if (jml == 0) { cout << BYELLOW << "  Belum ada produk.\n" << RESET;
return; }
    selectionSortHargaAsc(tmp, jml);
    try {
        string inp;
        cout << BGREEN << "  Masukkan harga (Rp): " << BHARGA;
        getline(cin, inp); cout << RESET;
        double target = bacaHarga(inp);
        int low = 0, high = jml - 1, posisi = -1;
        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (tmp[mid].harga.satuan == target) { posisi = mid; break; }
            else if (tmp[mid].harga.satuan < target) low = mid + 1;
            else high = mid - 1;
        }
        if (posisi == -1) {
            cout << BRED << "  Harga " << rupiahFormat(target) << " tidak
ada.\n" << RESET;
            double selMin = -1, hTerdekat = 0;
            for (int i = 0; i < jml; i++) {
                double s = tmp[i].harga.satuan - target; if (s<0) s=-s;
                if (selMin<0||s<selMin) { selMin=s;
hTerdekat=tmp[i].harga.satuan; }
            }
            cout << BYELLOW << "  Produk harga terdekat (" << BHARGA <<
rupiahFormat(hTerdekat) << BYELLOW << "):\n" << RESET;
            cetakHeaderTabel();
            int no = 0;
            for (int i = 0; i < jml; i++)
                if (tmp[i].harga.satuan == hTerdekat)
cetakBarisProdukTabel(++no, tmp[i]);
```

```
        } else {
            cetakHeaderTabel();
            int no = 0;
            for (int i = 0; i < jml; i++)
                if (tmp[i].harga.satuan == target)
cetakBarisProdukTabel(++no, tmp[i]);
            cout << BGREEN << "  [OK] Ditemukan " << no << " produk.\n" <<
RESET;
        }
        cetakFooterTabel();
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const out_of_range& e)       { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
}

// MENU LIHAT PRODUK
void menuKelolaLihatProduk() {
    string p;
    while (true) {
        cetakHeader("TAMPILAN PRODUK", BCYAN);
        cout << BYELLOW << "  -- Lihat --\n" << RESET;
        cout << BMAGENTA << "  [1]" << BWHITE << " Semua Produk (Default)\n"
<< RESET;
        cout << BYELLOW << "  -- Urutkan --\n" << RESET;
        cout << BMAGENTA << "  [2]" << BWHITE << " Nama Z -> A\n" << RESET;
        cout << BMAGENTA << "  [3]" << BWHITE << " Harga Termurah ->
Termahal\n" << RESET;
        cout << BMAGENTA << "  [4]" << BWHITE << " Stok Sedikit ->
Terbanyak\n" << RESET;
        cout << BYELLOW << "  -- Cari --\n" << RESET;
        cout << BMAGENTA << "  [5]" << BWHITE << " Cari by Jenis\n" <<
RESET;
        cout << BMAGENTA << "  [6]" << BWHITE << " Cari by Harga\n" <<
RESET;
        cout << BRED     << "  [0]" << BWHITE << " Kembali\n" << RESET;
        cetakGaris(BCYAN);
        cout << BGREEN << "  Pilih: " << BYELLOW; getline(cin, p); cout <<
RESET;
        if      (p=="1") { cout << BYELLOW << "\n  [Semua Produk]\n" <<
RESET; tampilTabel(0); }
        else if (p=="2") { cout << BYELLOW << "\n  [Nama Z -> A]\n" <<
RESET; tampilDenganSorting(1); }
        else if (p=="3") { cout << BYELLOW << "\n  [Harga Termurah]\n" <<
```

```
RESET; tampilDenganSorting(2); }
        else if (p=="4") { cout << BYELLOW << "\n  [Stok Sedikit]\n" <<
RESET; tampilDenganSorting(3); }
        else if (p=="5") cariNama();
        else if (p=="6") cariHarga();
        else if (p=="0") return;
        else cout << BRED << "  [!] Pilihan tidak valid.\n" << RESET;
    }
}

// KERANJANG BELANJA
void tampilKeranjang(ItemKeranjang keranjang[], int jmlKeranjang) {
    cetakHeader("KERANJANG BELANJA", BMAGENTA);
    cout << BMAGENTA <<
"+-----+----------------------+-------+----------------+-----------------+\
n";
    cout << "| " << BYELLOW  << left  << setw(3)  << "No"
         << BMAGENTA << " | " << BGREEN   << left  << setw(20) << "Nama
Produk"
         << BMAGENTA << " | " << BCYAN    << right << setw(5)  << "Qty"
         << BMAGENTA << " | " << BHARGA   << right << setw(15) << "Harga
Satuan"
         << BMAGENTA << " | " << BGREEN   << right << setw(15) << "Subtotal"
         << BMAGENTA << " |\n";
    cout <<
"+-----+----------------------+-------+----------------+-----------------+\
n" << RESET;

    double grandTotal = 0;
    for (int i = 0; i < jmlKeranjang; i++) {
        Produk &p = daftarProduk[keranjang[i].idxProduk];
        double sub = p.harga.satuan * keranjang[i].jumlah;
        grandTotal += sub;
        string nama = p.nama;
        if ((int)nama.size() > 20) nama = nama.substr(0, 17) + "...";
        string hargaStr = rupiahFormat(p.harga.satuan);
        if ((int)hargaStr.size() > 15) hargaStr = hargaStr.substr(0, 12) +
"...";
        string subStr = rupiahFormat(sub);
        if ((int)subStr.size() > 15) subStr = subStr.substr(0, 12) + "...";

        cout << BMAGENTA << "| " << BYELLOW  << left  << setw(3)  << (i + 1)
             << BMAGENTA << " | " << BWHITE   << left  << setw(20) << nama
             << BMAGENTA << " | " << BCYAN    << right << setw(5)  <<
```

```
keranjang[i].jumlah
             << BMAGENTA << " | " << BHARGA   << right << setw(15) <<
hargaStr
             << BMAGENTA << " | " << BGREEN   << right << setw(15) << subStr
             << BMAGENTA << " |\n" << RESET;
    }
    cout << BMAGENTA <<
"+-----+----------------------+-------+-----------------+-----------------+\
n" << RESET;
    cout << BOLD << BWHITE << "  TOTAL: " << BHARGA <<
rupiahFormat(grandTotal) << RESET << "\n";
    cetakGaris(BMAGENTA);
}

void kurangiStok(int *s, int jumlah) {
    if (!s) throw runtime_error("Pointer stok null!");
    if (jumlah <= 0) throw invalid_argument("Jumlah harus > 0.");
    if (*s < jumlah) throw out_of_range("Stok tidak mencukupi!");
    *s -= jumlah;
}

int tampilProdukMerk(string merk, int peta[], int modeSort) {
    Produk tmp[100]; int idxAsli[100]; int jml = 0;
    for (int i = 0; i < totalProduk; i++) {
        if (!daftarProduk[i].aktif) continue;
        string md = daftarProduk[i].merk, mc = merk;
        for (int j = 0; j < (int)md.size(); j++) md[j] = tolower(md[j]);
        for (int j = 0; j < (int)mc.size(); j++) mc[j] = tolower(mc[j]);
        if (md != mc) continue;
        tmp[jml] = daftarProduk[i]; idxAsli[jml] = i; jml++;
    }
    if (modeSort==1) bubbleSortNamaDesc(tmp, jml);
    else if (modeSort==2) selectionSortHargaAsc(tmp, jml);
    else if (modeSort==3) insertionSortStokAsc(tmp, jml);
    for (int i = 0; i < jml; i++)
        for (int k = 0; k < totalProduk; k++)
            if (daftarProduk[k].nama==tmp[i].nama &&
daftarProduk[k].merk==tmp[i].merk && daftarProduk[k].aktif)
                { idxAsli[i]=k; break; }
    string label = "KATALOG: " + merk;
    cetakHeader(label, BCYAN);
    if      (modeSort==1) cout << BCYAN << "  [Urutan: Nama Z -> A]\n" <<
RESET;
    else if (modeSort==2) cout << BCYAN << "  [Urutan: Harga Termurah]\n" <<
```

```
RESET;
    else if (modeSort==3) cout << BCYAN << "  [Urutan: Stok Sedikit]\n" <<
RESET;
    cetakHeaderTabel();
    for (int i = 0; i < jml; i++) {
        peta[i] = idxAsli[i];
        cetakBarisProdukTabel(i + 1, tmp[i]);
    }
    if (jml == 0) {
        cout << BCYAN << "|  " << BYELLOW << left << setw(86) << "(belum ada
produk untuk merk ini)"
             << BCYAN << "|\n" << RESET;
    }
    cetakFooterTabel();
    return jml;
}

int menuSortingPelanggan() {
    string p;
    while (true) {
        cetakHeader("PILIH URUTAN TAMPILAN", BBLUE);
        cout << BMAGENTA << "  [1]" << BWHITE << " Pilih merek &
Keranjang\n" << RESET;
        cout << BMAGENTA << "  [2]" << BWHITE << " Harga Termurah ->
Termahal\n" << RESET;
        cout << BRED     << "  [0]" << BWHITE << " Kembali\n" << RESET;
        cetakGaris(BBLUE);
        cout << BGREEN << "  Pilih: " << BYELLOW; getline(cin, p); cout <<
RESET;
        if      (p=="1") { cout << BCYAN << "  -> Nama Z -> A\n" << RESET;
return 1; }
        else if (p=="2") { cout << BCYAN << "  -> Harga Termurah\n" <<
RESET; return 2; }
        else if (p=="0") return -1;
        else cout << BRED << "  [!] Tidak valid!\n" << RESET;
    }
}

void tambahKeKeranjangDariMerk(ItemKeranjang keranjang[], int &jmlKeranjang,
int peta[], int no) {
    try {
        if (jmlKeranjang >= MAKS_KERANJANG)
            throw length_error("Keranjang penuh!");
        string pp, pj;
```

```cpp
        cout << BGREEN << "  Nomor produk (0=batal): " << BYELLOW;
        getline(cin, pp); cout << RESET;
        if (pp=="0") { cout << BYELLOW << "  Dibatalkan.\n" << RESET;
return; }
        int npm = bacaAngka(pp);
        if (npm<1||npm>no) throw out_of_range("Nomor di luar jangkauan.");
        int idx = peta[npm-1];
        Produk &p = daftarProduk[idx];
        int diKeranjang = 0;
        for (int i=0;i<jmlKeranjang;i++) if(keranjang[i].idxProduk==idx)
diKeranjang+=keranjang[i].jumlah;
        int tersedia = p.stok.jumlah - diKeranjang;
        if (tersedia<=0) throw out_of_range("Stok habis di keranjang!");
        cout << BGREEN << "  Jumlah '" << BWHITE << p.nama << BGREEN << "'
(stok: " << tersedia << "): " << BYELLOW;
        getline(cin, pj); cout << RESET;
        int jml = bacaAngka(pj);
        if (jml<=0) throw invalid_argument("Jumlah harus > 0.");
        if (jml>tersedia) throw out_of_range("Melebihi stok tersedia!");
        int ada=0;
        for (int i=0;i<jmlKeranjang;i++)
if(keranjang[i].idxProduk==idx){keranjang[i].jumlah+=jml;ada=1;break;}
        if (!ada) {keranjang[jmlKeranjang].idxProduk=idx;
keranjang[jmlKeranjang].jumlah=jml; jmlKeranjang++;}
        cout << BGREEN << "  [OK] " << p.nama << " x" << jml << " masuk
keranjang!\n" << RESET;
    } catch (const length_error& e)    { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const invalid_argument& e)  { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const out_of_range& e)      { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
}

// STRUK BELANJA
#define STRUK_W      62
#define STRUK_INNER  60
#define STRUK_SEP
"+------------------------------------------------------------+"
#define STRUK_ITEM_W 24   // lebar kolom nama produk di struk
#define STRUK_QTY_W   5   // lebar kolom qty
#define STRUK_SUB_W  16   // lebar kolom subtotal

void cetakStrukBaris(string label, string value, string labelColor, string
```

```
valueColor) {
    int valueW = STRUK_INNER - 4 - 16;  // 4 = "| " + " |", 16 = label(14) +
": "
    cout << BCYAN << "| " << RESET
         << labelColor << left << setw(14) << label << RESET
         << BCYAN << ": " << RESET
         << valueColor << left << setw(valueW) << value << RESET
         << BCYAN << " |\n" << RESET;
}

void prosesCheckout(ItemKeranjang keranjang[], int &jmlKeranjang, string
namaUser) {
    if (jmlKeranjang==0) { cout << BRED << "  [!] Keranjang kosong!\n" <<
RESET; return; }
    tampilKeranjang(keranjang, jmlKeranjang);
    string k;
    while (true) {
        cout << BYELLOW << "  Lanjut bayar? (y/n): " << BWHITE;
        getline(cin, k); cout << RESET;
        if (k=="y"||k=="Y"||k=="n"||k=="N") break;
        cout << BRED << "  [!] Input tidak valid!\n" << RESET;
    }
    if (k!="y"&&k!="Y") { cout << BYELLOW << "  Checkout dibatalkan.\n" <<
RESET; return; }
    try {
        string metode="", status="", pm;
        double bayar=0, kembali=0; int adaKembali=0;
        cout << "\n" << BYELLOW << "  -- Metode Pembayaran --\n" << RESET;
        cout << BMAGENTA << "  [1]" << BWHITE << " Tunai\n" << RESET;
        cout << BMAGENTA << "  [2]" << BWHITE << " Transfer Bank\n" <<
RESET;
        cout << BMAGENTA << "  [3]" << BWHITE << " QRIS\n" << RESET;
        cout << BMAGENTA << "  [4]" << BWHITE << " Kartu Debit/Kredit\n" <<
RESET;
        cout << BGREEN << "  Pilih: " << BYELLOW; getline(cin, pm); cout <<
RESET;
        double total=0;
        for (int i=0;i<jmlKeranjang;i++)
            total += daftarProduk[keranjang[i].idxProduk].harga.satuan *
keranjang[i].jumlah;
        if (pm=="1") {
            metode="Tunai"; string ib;
            cout << BGREEN << "  Jumlah bayar (Rp): " << BHARGA;
getline(cin, ib); cout << RESET;
```

```
            bayar = bacaHarga(ib);
            if (bayar<total) throw runtime_error("Uang kurang " +
rupiahFormat(total-bayar) + "!");
            kembali=bayar-total; adaKembali=1; status="LUNAS";
        } else if (pm=="2") { metode="Transfer Bank"; status="LUNAS
(Transfer)"; }
        else if (pm=="3")   { metode="QRIS";          status="LUNAS (QRIS)";
}
        else if (pm=="4")   { metode="Kartu Debit/Kredit"; status="LUNAS
(Kartu)"; }
        else throw invalid_argument("Metode tidak valid.");
        for (int i=0;i<jmlKeranjang;i++)
            kurangiStok(&daftarProduk[keranjang[i].idxProduk].stok.jumlah,
keranjang[i].jumlah);

        string wt = getWaktuSekarang(), nt = generateNoTransaksi();

// CETAK STRUK
        cout << "\n";
        cout << BGREEN << STRUK_SEP << "\n" << RESET;

        {
            string judul = "STRUK PEMBELIAN - GLOW UP STORE";
            int pad = (STRUK_INNER - (int)judul.size()) / 2;
            int padR = STRUK_INNER - (int)judul.size() - pad;
            cout << BGREEN << "|";
            for(int i=0;i<pad;i++) cout<<" ";
            cout << BOLD << BWHITE << judul << RESET << BGREEN;
            for(int i=0;i<padR;i++) cout<<" ";
            cout << "|\n" << RESET;
        }
        cout << BGREEN << STRUK_SEP << "\n" << RESET;

        cetakStrukBaris("No. Transaksi", nt,       BCYAN,    BWHITE);
        cetakStrukBaris("Waktu",         wt,       BCYAN,    BWHITE);
        cetakStrukBaris("Pelanggan",     namaUser, BCYAN,    BMAGENTA);

        cout << BGREEN << STRUK_SEP << "\n" << RESET;

        #define SW_NAMA 28
        #define SW_QTY   5
        #define SW_SUB  17

        cout << BCYAN << "+";
```

```
        for(int i=0;i<SW_NAMA+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_QTY+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_SUB+2;i++) cout<<"-";
        cout << "+\n" << RESET;

        {
            string hP="Produk", hQ="Qty", hS="Subtotal";
            int pPL=(SW_NAMA-(int)hP.size())/2,
pPR=SW_NAMA-(int)hP.size()-pPL;
            int pQL=(SW_QTY-(int)hQ.size())/2,
pQR=SW_QTY-(int)hQ.size()-pQL;
            int pSL=(SW_SUB-(int)hS.size())/2,
pSR=SW_SUB-(int)hS.size()-pSL;
            cout << BCYAN << "| " << BGREEN;
            for(int i=0;i<pPL;i++) cout<<" ";
            cout<<hP;
            for(int i=0;i<pPR;i++) cout<<" ";
            cout << BCYAN << " | " << BYELLOW;
            for(int i=0;i<pQL;i++) cout<<" ";
            cout<<hQ;
            for(int i=0;i<pQR;i++) cout<<" ";
            cout << BCYAN << " | " << BHARGA;
            for(int i=0;i<pSL;i++) cout<<" ";
            cout<<hS;
            for(int i=0;i<pSR;i++) cout<<" ";
            cout << BCYAN << " |\n" << RESET;
        }
        cout << BCYAN << "+";
        for(int i=0;i<SW_NAMA+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_QTY+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_SUB+2;i++) cout<<"-";
        cout << "+\n" << RESET;

        int totalQty = 0;
        for (int i=0;i<jmlKeranjang;i++) {
            Produk &p=daftarProduk[keranjang[i].idxProduk];
            double sub=p.harga.satuan*keranjang[i].jumlah;
            totalQty += keranjang[i].jumlah;
            string nama=p.nama;
            if((int)nama.size()>SW_NAMA)
```

```
nama=nama.substr(0,SW_NAMA-3)+"...";
            string subStr=rupiahFormat(sub);
            if((int)subStr.size()>SW_SUB)
subStr=subStr.substr(0,SW_SUB-3)+"...";

            cout << BCYAN << "| " << RESET
                 << BWHITE << left  << setw(SW_NAMA) << nama  << RESET
                 << BCYAN << " | " << RESET
                 << BYELLOW << right << setw(SW_QTY)  << keranjang[i].jumlah
<< RESET
                 << BCYAN << " | " << RESET
                 << BHARGA << right << setw(SW_SUB)  << subStr << RESET
                 << BCYAN << " |\n" << RESET;
        }
        cout << BCYAN << "+";
        for(int i=0;i<SW_NAMA+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_QTY+2;i++) cout<<"-";
        cout << "+";
        for(int i=0;i<SW_SUB+2;i++) cout<<"-";
        cout << "+\n" << RESET;
        {
            string totalStr = rupiahFormat(total);
            if((int)totalStr.size()>SW_SUB)
totalStr=totalStr.substr(0,SW_SUB-3)+"...";
            cout << BCYAN << "| " << RESET
                 << BOLD << BWHITE  << left  << setw(SW_NAMA) << "TOTAL" <<
RESET
                 << BCYAN << " | " << RESET
                 << BOLD << BYELLOW << right << setw(SW_QTY)  << totalQty <<
RESET
                 << BCYAN << " | " << RESET
                 << BOLD << BHARGA  << right << setw(SW_SUB)  << totalStr <<
RESET
                 << BCYAN << " |\n" << RESET;
        }

        cout << BGREEN << STRUK_SEP << "\n" << RESET;

// Dibayar & Kembalian (hanya tunai)
        if (adaKembali) {
            string bayarStr   = rupiahFormat(bayar);
            string kembaliStr = rupiahFormat(kembali);
            cetakStrukBaris("Dibayar",   bayarStr,   BCYAN, BHARGA);
```

```cpp
            cetakStrukBaris("Kembalian", kembaliStr, BCYAN, BGREEN);
            cout << BGREEN << STRUK_SEP << "\n" << RESET;
        }

        cetakStrukBaris("Metode", metode, BCYAN, BMAGENTA);
        cetakStrukBaris("Status", status, BCYAN, BGREEN);
        cout << BGREEN << STRUK_SEP << "\n" << RESET;

// Pesan penutup tengah
        {
            string pesan = "Makasih udah belanja! Semoga cocok :)";
            int pad  = (STRUK_INNER - (int)pesan.size()) / 2;
            int padR = STRUK_INNER - (int)pesan.size() - pad;
            cout << BGREEN << "|";
            for(int i=0;i<pad;i++) cout<<" ";
            cout << BOLD << BMAGENTA << pesan << RESET << BGREEN;
            for(int i=0;i<padR;i++) cout<<" ";
            cout << "|\n" << RESET;
        }
        cout << BGREEN << STRUK_SEP << "\n" << RESET;

        jmlKeranjang=0;
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << "\n  Checkout dibatalkan.\n" << RESET; }
    catch (const runtime_error& e)      { cout << BRED << "  [!] " <<
e.what() << "\n  Checkout dibatalkan.\n" << RESET; }
    catch (const out_of_range& e)       { cout << BRED << "  [!] " <<
e.what() << "\n  Checkout dibatalkan.\n" << RESET; }
}

// MENU MERK + KERANJANG

void menuMerkDanKeranjang(string namaUser) {
    ItemKeranjang keranjang[50]; int jmlKeranjang=0;
    int modeSort = menuSortingPelanggan();
    if (modeSort==-1) return;
    string merkAktif=""; int peta[100]; int noProduk=0;
    string p;
    while (true) {
        cetakHeader("PILIH MERK & KERANJANG", BMAGENTA);
        cout << BCYAN << "  Urutan   : " << BWHITE;
        if (modeSort==1) cout << "Nama Z -> A\n";
        else if (modeSort==2) cout << "Harga Termurah\n";
        else cout << "Default\n";
```

```
        if (!merkAktif.empty())
            cout << BCYAN << "  Merk      : " << BWHITE << merkAktif << " (" << noProduk << " produk)\n";
        cout << BCYAN << "  Keranjang: " << BGREEN << jmlKeranjang << BCYAN << " item\n" << RESET;
        cout << "\n";
        cout << BYELLOW << "  -- Pilih Merk --\n" << RESET;
        cout << BMAGENTA << "  [1]" << BWHITE << " Wardah     " << BMAGENTA << "[2]" << BWHITE << " Glad2Glow\n";
        cout << BMAGENTA << "  [3]" << BWHITE << " Emina      " << BMAGENTA << "[4]" << BWHITE << " Originote\n";
        cout << BMAGENTA << "  [5]" << BWHITE << " Skintific\n" << RESET;
        cout << BYELLOW << "  -- Keranjang --\n" << RESET;
        cout << BMAGENTA << "  [6]" << BWHITE << " Lihat keranjang\n";
        cout << BMAGENTA << "  [7]" << BWHITE << " Hapus item\n";
        cout << BMAGENTA << "  [8]" << BWHITE << " Checkout\n";
        cout << BRED     << "  [0]" << BWHITE << " Kembali\n" << RESET;
        cetakGaris(BMAGENTA);
        cout << BGREEN << "  Pilih: " << BYELLOW; getline(cin, p); cout << RESET;
        string mb="";
        if (p=="1") mb="Wardah"; else if (p=="2") mb="Glad2Glow";
        else if (p=="3") mb="Emina"; else if (p=="4") mb="Originote";
        else if (p=="5") mb="Skintific";
        if (!mb.empty()) {
            merkAktif=mb;
            noProduk=tampilProdukMerk(merkAktif, peta, modeSort);
            if (noProduk>0) {
                string t;
                while(true) {
                    cout << BYELLOW << "  Tambah ke keranjang? (y/n): " << BWHITE;
                    getline(cin, t); cout << RESET;
                    if (t=="y"||t=="Y"||t=="n"||t=="N") break;
                    cout << BRED << "  [!] Input tidak valid!\n" << RESET;
                }
                if (t=="y"||t=="Y") tambahKeKeranjangDariMerk(keranjang, jmlKeranjang, peta, noProduk);
            }
            continue;
        }
        if (p=="6") {
            if (jmlKeranjang==0) { cout << BYELLOW << "  Keranjang kosong.\n" << RESET; continue; }
```

```
            tampilKeranjang(keranjang, jmlKeranjang); continue;
        }
        if (p=="7") {
            if (jmlKeranjang==0) { cout << BRED << "  [!] Keranjang
kosong!\n" << RESET; continue; }
            tampilKeranjang(keranjang, jmlKeranjang);
            try {
                string hs;
                cout << BGREEN << "  Hapus nomor berapa? " << BYELLOW;
getline(cin, hs); cout << RESET;
                int ih = bacaAngka(hs)-1;
                if (ih<0||ih>=jmlKeranjang) throw out_of_range("Nomor di
luar jangkauan.");
                string nh = daftarProduk[keranjang[ih].idxProduk].nama;
                string kh;
                while(true) {
                    cout << BYELLOW << "  Hapus '" << BWHITE << nh <<
BYELLOW << "'? (y/n): " << BWHITE;
                    getline(cin, kh); cout << RESET;
                    if (kh=="y"||kh=="Y"||kh=="n"||kh=="N") break;
                    cout << BRED << "  [!] Input tidak valid!\n" << RESET;
                }
                if (kh=="n"||kh=="N") { cout << BYELLOW << "  Dibatalkan.\n"
<< RESET; continue; }
                for (int i=ih;i<jmlKeranjang-1;i++)
keranjang[i]=keranjang[i+1];
                jmlKeranjang--;
                cout << BGREEN << "  [OK] '" << nh << "' dihapus dari
keranjang.\n" << RESET;
            } catch (const invalid_argument& e) { cout << BRED << "  [!] "
<< e.what() << RESET << "\n"; }
            catch (const out_of_range& e)      { cout << BRED << "  [!] "
<< e.what() << RESET << "\n"; }
            continue;
        }
        if (p=="8") { prosesCheckout(keranjang, jmlKeranjang, namaUser);
continue; }

        if (p=="0") return;
        cout << BRED << "  [!] Pilihan tidak valid!\n" << RESET;
    }
}

// LOGIN & REGISTER
```

```
void doRegister() {
    if (jumlahPengguna>=MAKS_PENGGUNA) { cout << BRED << "  [!] Slot
pengguna penuh!\n" << RESET; return; }
    cetakHeader("DAFTAR AKUN BARU", BCYAN);
    try {
        string nama, pass;
        cout << BCYAN << "  Nama     : " << BWHITE; getline(cin, nama);
        if (nama.empty()) throw invalid_argument("Nama tidak boleh
kosong.");
        for (int i=0;i<jumlahPengguna;i++)
            if (dataPengguna[i].nama==nama) throw runtime_error("Nama sudah
dipakai!");
        cout << BCYAN << "  Password : " << BWHITE; getline(cin, pass); cout
<< RESET;
        if (pass.empty()) throw invalid_argument("Password tidak boleh
kosong.");
        dataPengguna[jumlahPengguna++]={nama, pass, "pelanggan", 1};
        simpanPengguna();
        cout << BGREEN << "  [OK] Akun '" << nama << "' berhasil dibuat!\n"
<< RESET;
    } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
    catch (const runtime_error& e)      { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; }
}

int doLogin(Pengguna* data, int jml) {
    int coba=0;
    while (coba<3) {
        cetakHeader("LOGIN GLOW UP STORE", BGREEN);
        try {
            string n, pw;
            cout << BCYAN << "  Nama     : " << BWHITE; getline(cin, n);
            cout << BCYAN << "  Password : " << BWHITE; getline(cin, pw);
cout << RESET;
            if (n.empty()||pw.empty()) throw invalid_argument("Nama dan
password tidak boleh kosong.");
            int idx=-1;
            for (int i=0;i<jml;i++)
                if (data[i].aktif&&data[i].nama==n&&data[i].password==pw) {
idx=i; break; }
            if (idx!=-1) {
                cout << BGREEN << "\n  [OK] Login berhasil! Halo, " <<
```

```
BMAGENTA << data[idx].nama << BGREEN << "!\n" << RESET;
                simpanRiwayatLogin(data[idx].nama, data[idx].role,
"BERHASIL");
                if (data[idx].role=="admin") menuAdmin(data[idx].nama);
                else menuPelanggan(data[idx].nama);
                return 1;
            }
            coba++;
            simpanRiwayatLogin(n, "-", "GAGAL");
            throw runtime_error("Nama atau password salah.");
        } catch (const invalid_argument& e) { cout << BRED << "  [!] " <<
e.what() << RESET << "\n"; coba++; }
        catch (const runtime_error& e) {
            cout << BRED << "\n  [!] " << e.what() << RESET << "\n";
            if (coba<3) cout << BYELLOW << "  Sisa percobaan: " << (3-coba)
<< RESET << "\n";
        }
    }
    cout << BRED << "\n  [!] Terlalu banyak percobaan.\n" << RESET;
    return 0;
}

// MENU ADMIN & PELANGGAN

void menuAdmin(string namaUser) {
    string p;
    while (true) {
        cetakHeader("GLOW UP STORE - ADMIN", BYELLOW);
        cout << BYELLOW << "  Halo, " << BMAGENTA << namaUser << BYELLOW <<
"!\n\n" << RESET;
        cout << BMAGENTA << "  [1]" << BWHITE << " Tambah Produk\n";
        cout << BMAGENTA << "  [2]" << BWHITE << " Edit Produk\n";
        cout << BMAGENTA << "  [3]" << BWHITE << " Hapus Produk\n";
        cout << BMAGENTA << "  [4]" << BWHITE << " Lihat Produk\n";
        cout << BMAGENTA << "  [5]" << BWHITE << " Riwayat Login\n" <<
RESET;
        cout << BRED     << "  [0]" << BWHITE << " Logout\n" << RESET;
        cetakGaris(BYELLOW);
        cout << BGREEN << "  Pilih (0-5): " << BYELLOW; getline(cin, p);
cout << RESET;
        if      (p=="1") tambahProduk();
        else if (p=="2") editProduk();
        else if (p=="3") hapusProduk();
        else if (p=="4") menuKelolaLihatProduk();
```

```
        else if (p=="5") { cetakHeader("RIWAYAT LOGIN", BCYAN);
tampilRiwayatLogin(); }
        else if (p=="0") { cout << BGREEN << "  Sampai jumpa!\n" << RESET;
return; }
        else cout << BRED << "  [!] Pilihan tidak valid.\n" << RESET;
    }
}

void menuPelanggan(string namaUser) {
    string p;
    while (true) {
        cetakHeader("GLOW UP STORE - PELANGGAN", BMAGENTA);
        cout << BMAGENTA << "  Halo, " << BCYAN << namaUser << BMAGENTA <<
"!\n\n" << RESET;
        cout << BMAGENTA << "  [1]" << BWHITE << " Pilih Merk Skincare\n";
        cout << BMAGENTA << "  [2]" << BWHITE << " Cari by Jenis\n" <<
RESET;
        cout << BRED     << "  [0]" << BWHITE << " Keluar\n" << RESET;
        cetakGaris(BMAGENTA);
        cout << BGREEN << "  Pilih (0-2): " << BYELLOW; getline(cin, p);
cout << RESET;
        if      (p=="1") menuMerkDanKeranjang(namaUser);
        else if (p=="2") cariNama();
        else if (p=="0") { cout << BGREEN << "  Makasih sudah mampir! Sampai
jumpa :)\n" << RESET; return; }
        else cout << BRED << "  [!] Pilihan tidak valid.\n" << RESET;
    }
}

// MAIN

int main() {
    loadPengguna();
    int adminAda=0, tamuAda=0;
    for (int i=0;i<jumlahPengguna;i++) {
        if (dataPengguna[i].nama=="admin") adminAda=1;
        if (dataPengguna[i].nama=="Tamu")  tamuAda=1;
    }
    if (!adminAda) dataPengguna[jumlahPengguna++]={"admin","123","admin",1};
    if (!tamuAda)
dataPengguna[jumlahPengguna++]={"Tamu","000","pelanggan",1};

// ===== WARDAH =====
    daftarProduk[totalProduk++]={"Lightening Serum",       "Wardah",
```

```
{55000, "IDR"},{40,"pcs"},"Serum",      1};
    daftarProduk[totalProduk++]={"Hydrating Toner",        "Wardah",
{38000, "IDR"},{30,"pcs"},"Toner",      1};
    daftarProduk[totalProduk++]={"Daily Moisturizer",      "Wardah",
{52000, "IDR"},{20,"pcs"},"Moisturizer",1};
    daftarProduk[totalProduk++]={"Face Wash Aloe",         "Wardah",
{30000, "IDR"},{25,"pcs"},"Face Wash",  1};
    daftarProduk[totalProduk++]={"UV Shield Sunscreen",    "Wardah",
{48000, "IDR"},{35,"pcs"},"Sunscreen",  1};
// ===== GLAD2GLOW =====
    daftarProduk[totalProduk++]={"Brightening Serum",
"Glad2Glow",{62000, "IDR"},{25,"pcs"},"Serum",      1};
    daftarProduk[totalProduk++]={"Exfo Toner AHA BHA",
"Glad2Glow",{57000, "IDR"},{20,"pcs"},"Toner",      1};
    daftarProduk[totalProduk++]={"Moisturizer Gel",
"Glad2Glow",{45000, "IDR"},{30,"pcs"},"Moisturizer",1};
    daftarProduk[totalProduk++]={"Gentle Face Wash",
"Glad2Glow",{35000, "IDR"},{20,"pcs"},"Face Wash",  1};
    daftarProduk[totalProduk++]={"Daily Sun Protect
SPF35","Glad2Glow",{53000, "IDR"},{22,"pcs"},"Sunscreen",  1};
// ===== EMINA =====
    daftarProduk[totalProduk++]={"Acne Care Serum",        "Emina",
{42000, "IDR"},{28,"pcs"},"Serum",      1};
    daftarProduk[totalProduk++]={"Bright Stuff Toner",     "Emina",
{35000, "IDR"},{45,"pcs"},"Toner",      1};
    daftarProduk[totalProduk++]={"Ms Pimple Moisturizer",  "Emina",
{36000, "IDR"},{30,"pcs"},"Moisturizer",1};
    daftarProduk[totalProduk++]={"Bright Stuff Face Wash", "Emina",
{28000, "IDR"},{40,"pcs"},"Face Wash",  1};
    daftarProduk[totalProduk++]={"Sun Protection SPF30",   "Emina",
{40000, "IDR"},{50,"pcs"},"Sunscreen",  1};
 // ===== ORIGINOTE =====
    daftarProduk[totalProduk++]={"Niacinamide Serum",
"Originote",{72000, "IDR"},{20,"pcs"},"Serum",      1};
    daftarProduk[totalProduk++]={"Hyal Toner Essence",
"Originote",{65000, "IDR"},{18,"pcs"},"Toner",      1};
    daftarProduk[totalProduk++]={"Hyalucera Moisturizer",
"Originote",{75000, "IDR"},{22,"pcs"},"Moisturizer",1};
    daftarProduk[totalProduk++]={"Mild Cleanser FaceWash",
"Originote",{60000, "IDR"},{15,"pcs"},"Face Wash",  1};
    daftarProduk[totalProduk++]={"UV Barrier Sunscreen",
"Originote",{78000, "IDR"},{16,"pcs"},"Sunscreen",  1};
// ===== SKINTIFIC =====
    daftarProduk[totalProduk++]={"SymWhite Serum",
```

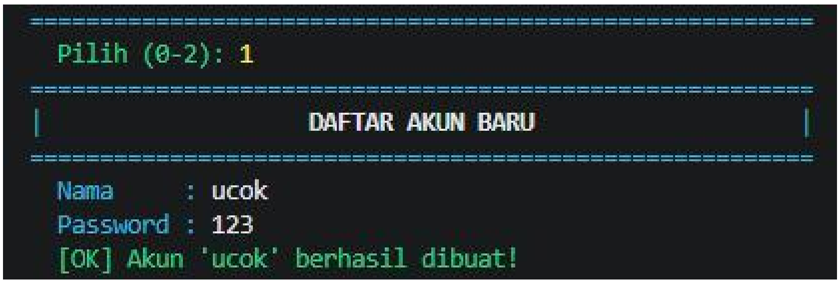
```
"Skintific",{125000,"IDR"},{12,"pcs"},"Serum",      1};
    daftarProduk[totalProduk++]={"AHA BHA PHA Toner",
"Skintific",{105000,"IDR"},{14,"pcs"},"Toner",      1};
    daftarProduk[totalProduk++]={"5x Ceramide Barrier",
"Skintific",{115000,"IDR"},{15,"pcs"},"Moisturizer",1};
    daftarProduk[totalProduk++]={"Acne Facial Wash",
"Skintific",{95000, "IDR"},{14,"pcs"},"Face Wash",  1};
    daftarProduk[totalProduk++]={"AMPM Sunscreen SPF50",
"Skintific",{110000,"IDR"},{13,"pcs"},"Sunscreen",  1};

    cout << "\n";
    cetakGaris(BMAGENTA);
    cetakTengah(BG_MAGENTA, BWHITE, "SELAMAT DATANG DI GLOW UP STORE!");
    cetakTengah(BG_MAGENTA, BWHITE, "Toko Skincare Terpercaya No.1");
    cetakGaris(BMAGENTA);
    cout << "\n";

    string pilihan; int berjalan=1;
    while (berjalan) {
        cetakHeader("MENU UTAMA", BCYAN);
        cout << BMAGENTA << "  [1]" << BWHITE << " Register\n";
        cout << BMAGENTA << "  [2]" << BWHITE << " Login\n" << RESET;
        cout << BRED     << "  [0]" << BWHITE << " Keluar\n" << RESET;
        cetakGaris(BCYAN);
        cout << BGREEN << "  Pilih (0-2): " << BYELLOW; getline(cin,
pilihan); cout << RESET;
        if      (pilihan=="1") doRegister();
        else if (pilihan=="2") { if (doLogin(dataPengguna,
jumlahPengguna)==0) berjalan=0; }
        else if (pilihan=="0") {
            cetakGaris(BMAGENTA);
            cetakTengah(BG_MAGENTA, BWHITE, "Terima kasih! Keep Glowing
Bestie!");
            cetakGaris(BMAGENTA);
            berjalan=0;
        }
        else cout << BRED << "  [!] Pilihan tidak valid.\n" << RESET;
    }
    return 0;
}
```

Gambar 1.5 Source Code Program

4. Hasil Output

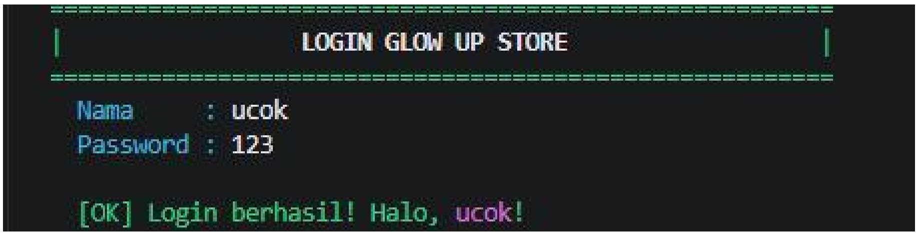
```
=============================================================
|               SELAMAT DATANG DI GLOW UP STORE!            |
|                 Toko Skincare Terpercaya No.1             |
=============================================================


=============================================================
|                         MENU UTAMA                        |
=============================================================
  [1] Register
  [2] Login
  [0] Keluar
=============================================================
  Pilih (0-2):
```

Gambar 1.6 Output Menu Utama

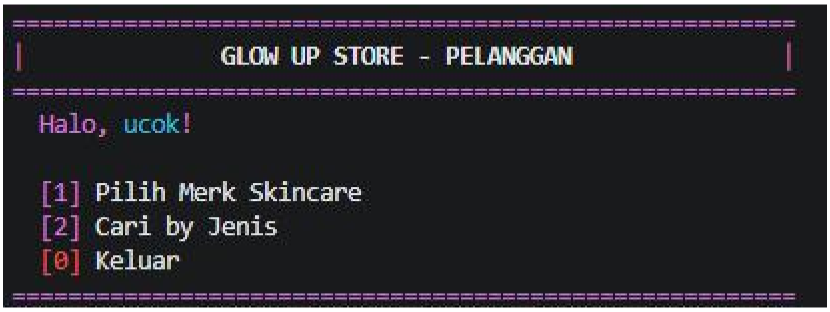


```
=============================================================
  Pilih (0-2): 1
=============================================================
|                      DAFTAR AKUN BARU                     |
=============================================================
  Nama     : ucok
  Password : 123
  [OK] Akun 'ucok' berhasil dibuat!
```

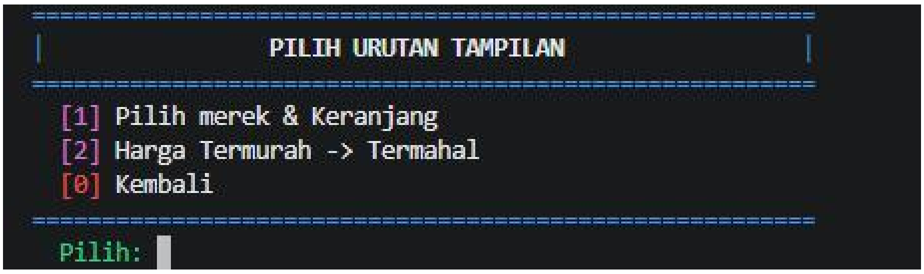
Gambar 1.7 Output Tampilan Registrasi/Pembuatan Akun Pelanggan



Gambar 1.8 Output Tampilan Menu Login Menu pelanggan

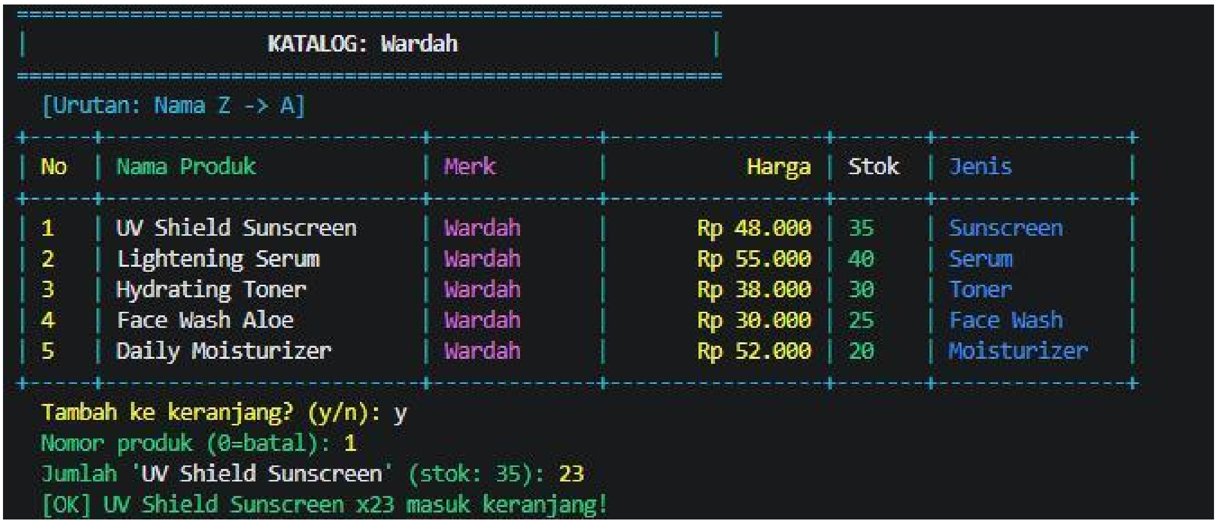


Gambar 1.9 Output Tampilan Halaman Menu Utama Pelanggan



Gambar 1.10 Output Tampilan Pilihan 1 Pada Menu Utama Pelanggan

Gambar 1.11 Output Tampilan Menu Pilih Merek Dan Keranjang Pelanggan



Gambar 1.12 Output Salah Satu contoh tampilan Katalog Produk Yang ada di Store Kami

Gambar 1.13 Output Tampilan Menu Keranjang Belanja Yang Ada Dalam Menu Pilih Merk dan keranjang

```
===========================================================
|                    KERANJANG BELANJA                    |
===========================================================
+-----+------------------------+-------+------------------+------------------+
| No  | Nama Produk            |  Qty  |    Harga Satuan  |         Subtotal |
+-----+------------------------+-------+------------------+------------------+
| 1   | UV Shield Sunscreen    |    23 |        Rp 48.000 |     Rp 1.104.000 |
+-----+------------------------+-------+------------------+------------------+
  TOTAL: Rp 1.104.000
===========================================================
  Hapus nomor berapa? 1
  Hapus 'UV Shield Sunscreen'? (y/n): y
  [OK] 'UV Shield Sunscreen' dihapus dari keranjang.
```

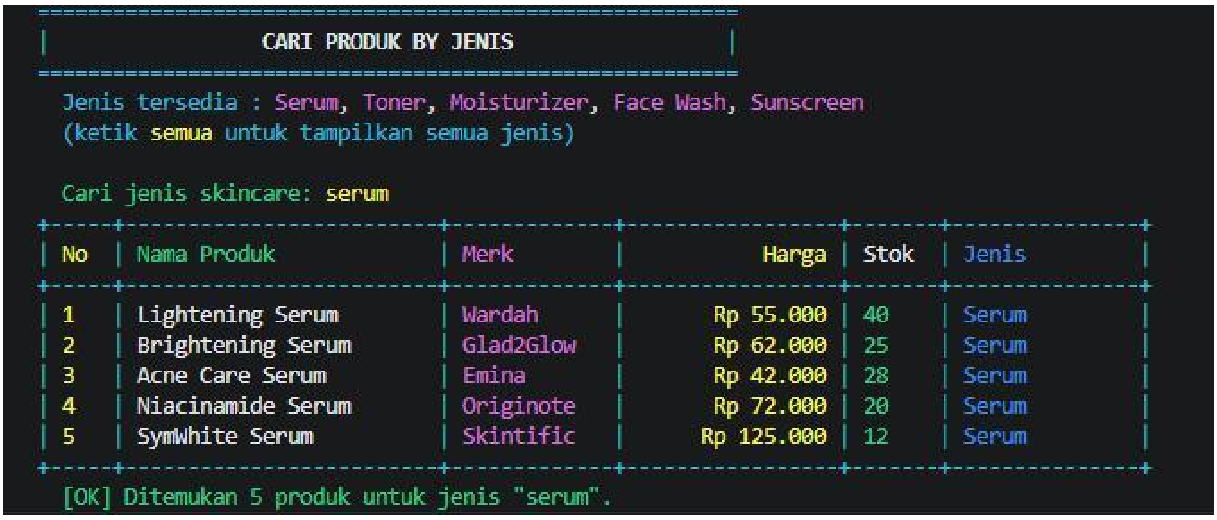
Gambar 1.14 Output Tampilan Fitur No 7 Yaitu,Untuk Mengahapus Produk Di Keranjang Kita

```
===========================================================
|                    KERANJANG BELANJA                    |
===========================================================
+-----+------------------------+-------+------------------+------------------+
| No  | Nama Produk            |  Qty  |    Harga Satuan  |         Subtotal |
+-----+------------------------+-------+------------------+------------------+
| 1   | UV Shield Sunscreen    |     3 |        Rp 48.000 |       Rp 144.000 |
+-----+------------------------+-------+------------------+------------------+
  TOTAL: Rp 144.000
===========================================================
  Lanjut bayar? (y/n): y

  -- Metode Pembayaran --
  [1] Tunai
  [2] Transfer Bank
  [3] QRIS
  [4] Kartu Debit/Kredit
  Pilih: 1
  Jumlah bayar (Rp): 200000
```

Gambar 1.15 Output Tampilan Fitur Cekout Beserta 4 Metode Pembayarannya

```
+-----------------------------------------------------------------+
|              STRUK PEMBELIAN - GLOW UP STORE                     |
+-----------------------------------------------------------------+
| No. Transaksi : TRX20260519215708                              |
| Waktu         : 19/05/2026 21:57:08                            |
| Pelanggan     : ucok                                           |
+-----------------------------------------------------------------+
+-------------------------------+--------+-------------------+
|            Produk             |  Qty   |     Subtotal      |
+-------------------------------+--------+-------------------+
| UV Shield Sunscreen           |      3 |        Rp 144.000 |
+-------------------------------+--------+-------------------+
| TOTAL                         |      3 |        Rp 144.000 |
+-----------------------------------------------------------------+
| Dibayar       : Rp 200.000                                     |
| Kembalian     : Rp 56.000                                      |
+-----------------------------------------------------------------+
| Metode        : Tunai                                          |
| Status        : LUNAS                                          |
+-----------------------------------------------------------------+
|          Makasih udah belanja! Semoga cocok :)                   |
+-----------------------------------------------------------------+
==========================================================
```

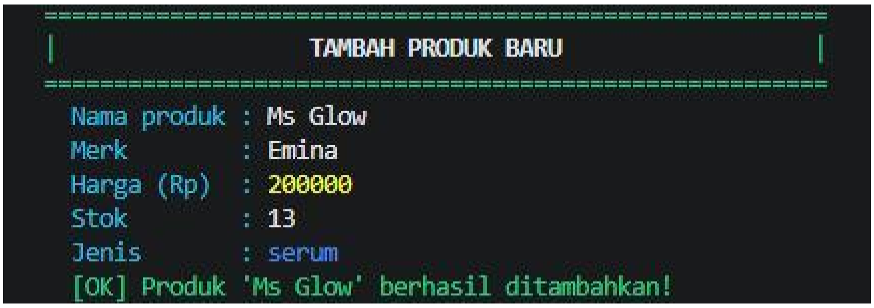
Gambar 1.16 Output Tampilan Struk Pembelian Pelanggan Yang Telah Berhasil Terbayar/Lunas



```
====================================================
|                CARI PRODUK BY JENIS               |
====================================================
  Jenis tersedia : Serum, Toner, Moisturizer, Face Wash, Sunscreen
  (ketik semua untuk tampilkan semua jenis)

  Cari jenis skincare: serum
+-----+-------------------------+-------------+------------------+-------+--------------+
| No  | Nama Produk             | Merk        |            Harga | Stok  | Jenis        |
+-----+-------------------------+-------------+------------------+-------+--------------+
| 1   | Lightening Serum        | Wardah      |        Rp 55.000 | 40    | Serum        |
| 2   | Brightening Serum       | Glad2Glow   |        Rp 62.000 | 25    | Serum        |
| 3   | Acne Care Serum         | Emina       |        Rp 42.000 | 28    | Serum        |
| 4   | Niacinamide Serum       | Originote   |        Rp 72.000 | 20    | Serum        |
| 5   | SymWhite Serum          | Skintific   |       Rp 125.000 | 12    | Serum        |
+-----+-------------------------+-------------+------------------+-------+--------------+
  [OK] Ditemukan 5 produk untuk jenis "serum".
```

Gambar 1.17 Output Salah Satu Contoh Tampilan Dari Menu Cari Produk By Jenis di Menu Utama Pelanggan

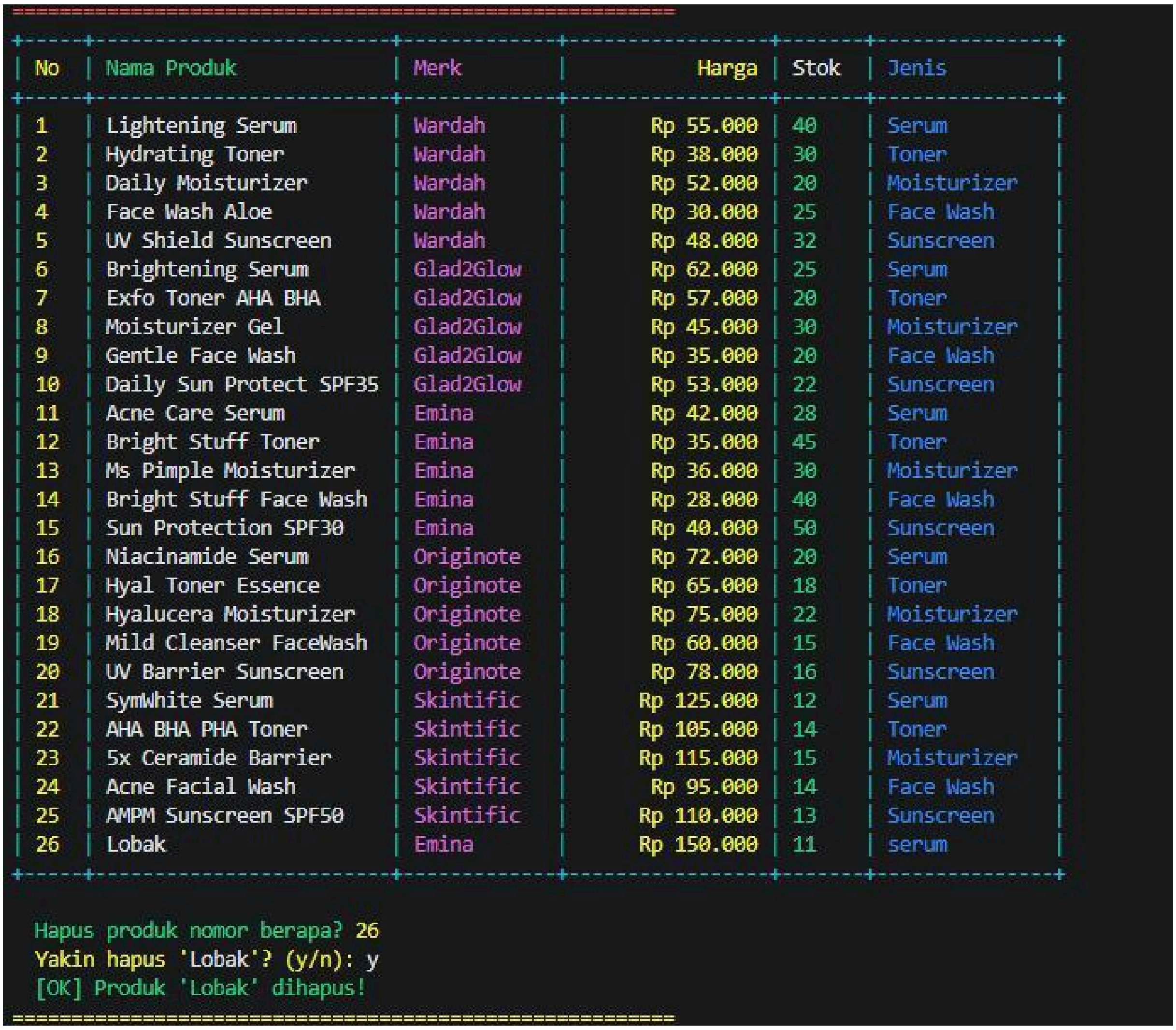
Gambar 1.18 Output Tampilan Menu Utama Admin



Gambar 1.19 Output Tampilan Fitur Untuk Menambahkan Produk Baru Pada Menu Admin

```
| 3    | Daily Moisturizer        | Wardah      |       Rp 52.000 | 20    | Moisturizer   |
| 4    | Face Wash Aloe           | Wardah      |       Rp 30.000 | 25    | Face Wash     |
| 5    | UV Shield Sunscreen      | Wardah      |       Rp 48.000 | 32    | Sunscreen     |
| 6    | Brightening Serum        | Glad2Glow   |       Rp 62.000 | 25    | Serum         |
| 7    | Exfo Toner AHA BHA       | Glad2Glow   |       Rp 57.000 | 20    | Toner         |
| 8    | Moisturizer Gel          | Glad2Glow   |       Rp 45.000 | 30    | Moisturizer   |
| 9    | Gentle Face Wash         | Glad2Glow   |       Rp 35.000 | 20    | Face Wash     |
| 10   | Daily Sun Protect SPF35  | Glad2Glow   |       Rp 53.000 | 22    | Sunscreen     |
| 11   | Acne Care Serum          | Emina       |       Rp 42.000 | 28    | Serum         |
| 12   | Bright Stuff Toner       | Emina       |       Rp 35.000 | 45    | Toner         |
| 13   | Ms Pimple Moisturizer    | Emina       |       Rp 36.000 | 30    | Moisturizer   |
| 14   | Bright Stuff Face Wash   | Emina       |       Rp 28.000 | 40    | Face Wash     |
| 15   | Sun Protection SPF30     | Emina       |       Rp 40.000 | 50    | Sunscreen     |
| 16   | Niacinamide Serum        | Originote   |       Rp 72.000 | 20    | Serum         |
| 17   | Hyal Toner Essence       | Originote   |       Rp 65.000 | 18    | Toner         |
| 18   | Hyalucera Moisturizer    | Originote   |       Rp 75.000 | 22    | Moisturizer   |
| 19   | Mild Cleanser FaceWash   | Originote   |       Rp 60.000 | 15    | Face Wash     |
| 20   | UV Barrier Sunscreen     | Originote   |       Rp 78.000 | 16    | Sunscreen     |
| 21   | SymWhite Serum           | Skintific   |      Rp 125.000 | 12    | Serum         |
| 22   | AHA BHA PHA Toner        | Skintific   |      Rp 105.000 | 14    | Toner         |
| 23   | 5x Ceramide Barrier      | Skintific   |      Rp 115.000 | 15    | Moisturizer   |
| 24   | Acne Facial Wash         | Skintific   |       Rp 95.000 | 14    | Face Wash     |
| 25   | AMPM Sunscreen SPF50     | Skintific   |      Rp 110.000 | 13    | Sunscreen     |
| 26   | Ms Glow                  | Emina       |      Rp 200.000 | 13    | serum         |
+------+--------------------------+-------------+-----------------+-------+---------------+

  Edit produk nomor berapa? 26

  Edit produk (Enter = tidak berubah)
  Nama  [Ms Glow]: Lobak
  Merk  [Emina]: Emina
  Harga [Rp 200.000]: 150000
  Stok  [13]: 11
  Jenis [serum]: serum
  [OK] Data berhasil diupdate!
==========================================================
```

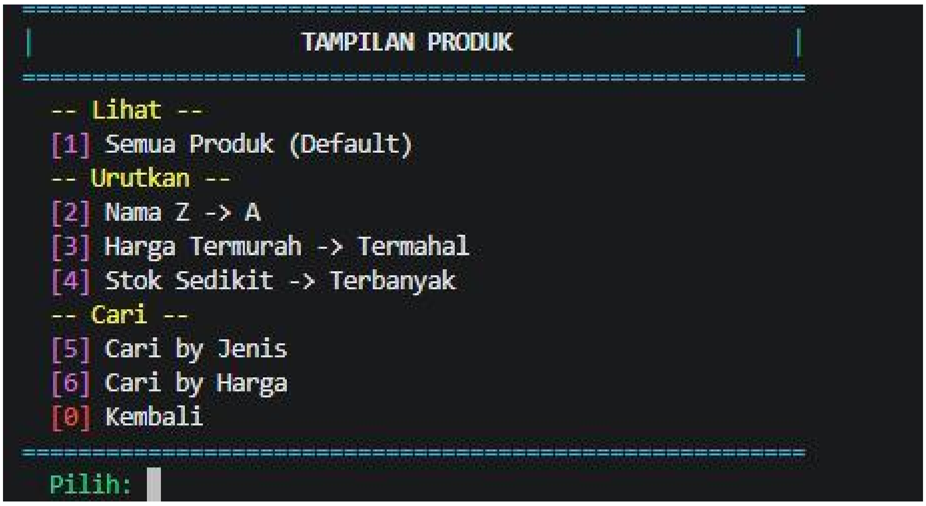
Gambar 1.20 Output Tampilan Fitur Ke 2 Yaitu Fitur Untuk Mengedit Nama,Merek,Harga,Stok,Dan Jenis Produk Tersebut



No	Nama Produk	Merk	Harga	Stok	Jenis
1	Lightening Serum	Wardah	Rp 55.000	40	Serum
2	Hydrating Toner	Wardah	Rp 38.000	30	Toner
3	Daily Moisturizer	Wardah	Rp 52.000	20	Moisturizer
4	Face Wash Aloe	Wardah	Rp 30.000	25	Face Wash
5	UV Shield Sunscreen	Wardah	Rp 48.000	32	Sunscreen
6	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
7	Exfo Toner AHA BHA	Glad2Glow	Rp 57.000	20	Toner
8	Moisturizer Gel	Glad2Glow	Rp 45.000	30	Moisturizer
9	Gentle Face Wash	Glad2Glow	Rp 35.000	20	Face Wash
10	Daily Sun Protect SPF35	Glad2Glow	Rp 53.000	22	Sunscreen
11	Acne Care Serum	Emina	Rp 42.000	28	Serum
12	Bright Stuff Toner	Emina	Rp 35.000	45	Toner
13	Ms Pimple Moisturizer	Emina	Rp 36.000	30	Moisturizer
14	Bright Stuff Face Wash	Emina	Rp 28.000	40	Face Wash
15	Sun Protection SPF30	Emina	Rp 40.000	50	Sunscreen
16	Niacinamide Serum	Originote	Rp 72.000	20	Serum
17	Hyal Toner Essence	Originote	Rp 65.000	18	Toner
18	Hyalucera Moisturizer	Originote	Rp 75.000	22	Moisturizer
19	Mild Cleanser FaceWash	Originote	Rp 60.000	15	Face Wash
20	UV Barrier Sunscreen	Originote	Rp 78.000	16	Sunscreen
21	SymWhite Serum	Skintific	Rp 125.000	12	Serum
22	AHA BHA PHA Toner	Skintific	Rp 105.000	14	Toner
23	5x Ceramide Barrier	Skintific	Rp 115.000	15	Moisturizer
24	Acne Facial Wash	Skintific	Rp 95.000	14	Face Wash
25	AMPM Sunscreen SPF50	Skintific	Rp 110.000	13	Sunscreen
26	Lobak	Emina	Rp 150.000	11	serum

Hapus produk nomor berapa? 26
Yakin hapus 'Lobak'? (y/n): y
[OK] Produk 'Lobak' dihapus!

Gambar 1.21 Output Tampilan Fitur Ke 3 Yaitu Fitur Untuk Menghapus Produk

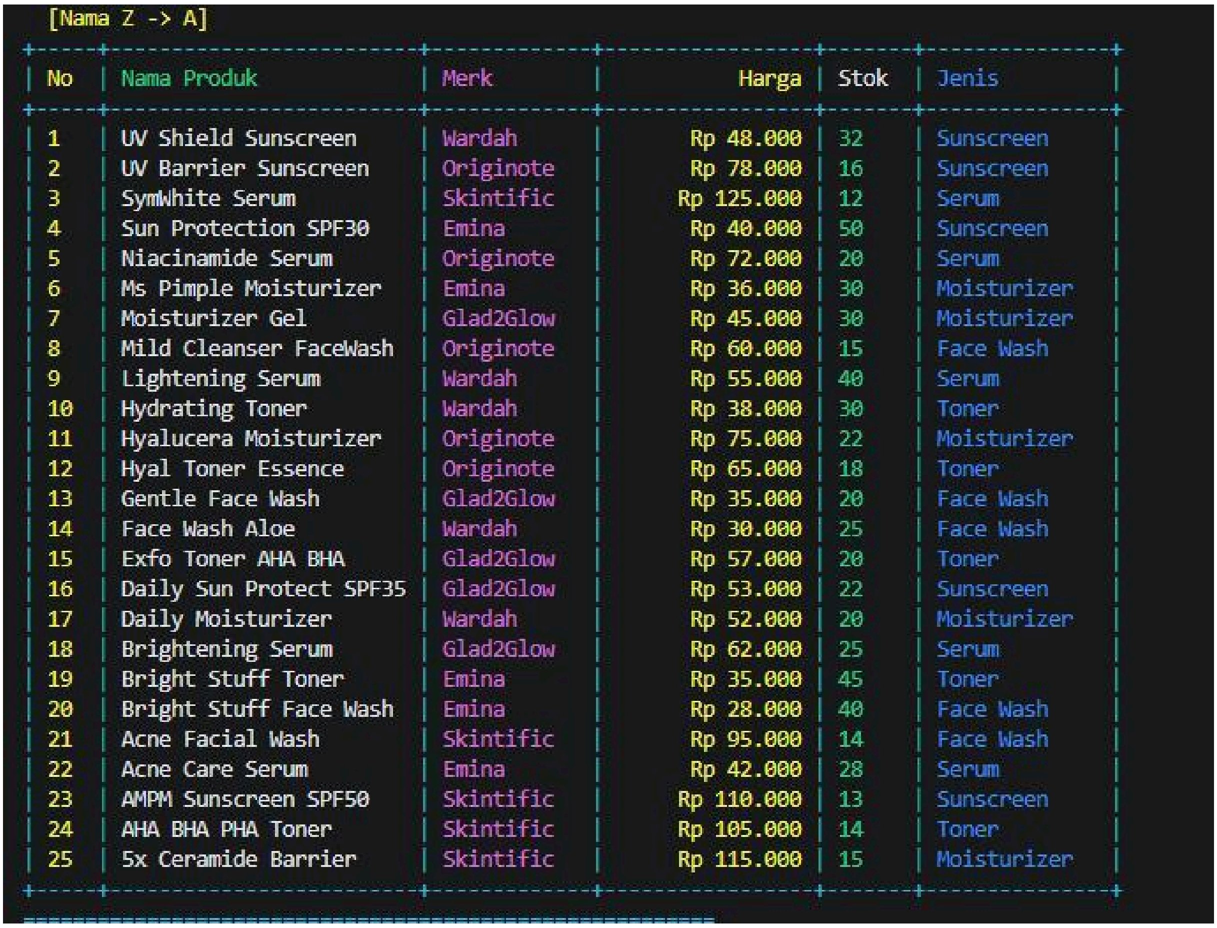


Gambar 1.22 Output Tampilan Menu Ke 4 Yaitu Menu Untuk Melihat Produk di Menu admin

[Semua Produk]

No	Nama Produk	Merk	Harga	Stok	Jenis
1	Lightening Serum	Wardah	Rp 55.000	40	Serum
2	Hydrating Toner	Wardah	Rp 38.000	30	Toner
3	Daily Moisturizer	Wardah	Rp 52.000	20	Moisturizer
4	Face Wash Aloe	Wardah	Rp 30.000	25	Face Wash
5	UV Shield Sunscreen	Wardah	Rp 48.000	32	Sunscreen
6	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
7	Exfo Toner AHA BHA	Glad2Glow	Rp 57.000	20	Toner
8	Moisturizer Gel	Glad2Glow	Rp 45.000	30	Moisturizer
9	Gentle Face Wash	Glad2Glow	Rp 35.000	20	Face Wash
10	Daily Sun Protect SPF35	Glad2Glow	Rp 53.000	22	Sunscreen
11	Acne Care Serum	Emina	Rp 42.000	28	Serum
12	Bright Stuff Toner	Emina	Rp 35.000	45	Toner
13	Ms Pimple Moisturizer	Emina	Rp 36.000	30	Moisturizer
14	Bright Stuff Face Wash	Emina	Rp 28.000	40	Face Wash
15	Sun Protection SPF30	Emina	Rp 40.000	50	Sunscreen
16	Niacinamide Serum	Originote	Rp 72.000	20	Serum
17	Hyal Toner Essence	Originote	Rp 65.000	18	Toner
18	Hyalucera Moisturizer	Originote	Rp 75.000	22	Moisturizer
19	Mild Cleanser FaceWash	Originote	Rp 60.000	15	Face Wash
20	UV Barrier Sunscreen	Originote	Rp 78.000	16	Sunscreen
21	SymWhite Serum	Skintific	Rp 125.000	12	Serum
22	AHA BHA PHA Toner	Skintific	Rp 105.000	14	Toner
23	5x Ceramide Barrier	Skintific	Rp 115.000	15	Moisturizer
24	Acne Facial Wash	Skintific	Rp 95.000	14	Face Wash
25	AMPM Sunscreen SPF50	Skintific	Rp 110.000	13	Sunscreen

Gambar 1.23 Output Tampilan Pertama Yaitu
Untuk Menampilan Semua Produk yang kami
miliki di Store (Menu Admin)



[Nama Z -> A]

No	Nama Produk	Merk	Harga	Stok	Jenis
1	UV Shield Sunscreen	Wardah	Rp 48.000	32	Sunscreen
2	UV Barrier Sunscreen	Originote	Rp 78.000	16	Sunscreen
3	SymWhite Serum	Skintific	Rp 125.000	12	Serum
4	Sun Protection SPF30	Emina	Rp 40.000	50	Sunscreen
5	Niacinamide Serum	Originote	Rp 72.000	20	Serum
6	Ms Pimple Moisturizer	Emina	Rp 36.000	30	Moisturizer
7	Moisturizer Gel	Glad2Glow	Rp 45.000	30	Moisturizer
8	Mild Cleanser FaceWash	Originote	Rp 60.000	15	Face Wash
9	Lightening Serum	Wardah	Rp 55.000	40	Serum
10	Hydrating Toner	Wardah	Rp 38.000	30	Toner
11	Hyalucera Moisturizer	Originote	Rp 75.000	22	Moisturizer
12	Hyal Toner Essence	Originote	Rp 65.000	18	Toner
13	Gentle Face Wash	Glad2Glow	Rp 35.000	20	Face Wash
14	Face Wash Aloe	Wardah	Rp 30.000	25	Face Wash
15	Exfo Toner AHA BHA	Glad2Glow	Rp 57.000	20	Toner
16	Daily Sun Protect SPF35	Glad2Glow	Rp 53.000	22	Sunscreen
17	Daily Moisturizer	Wardah	Rp 52.000	20	Moisturizer
18	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
19	Bright Stuff Toner	Emina	Rp 35.000	45	Toner
20	Bright Stuff Face Wash	Emina	Rp 28.000	40	Face Wash
21	Acne Facial Wash	Skintific	Rp 95.000	14	Face Wash
22	Acne Care Serum	Emina	Rp 42.000	28	Serum
23	AMPM Sunscreen SPF50	Skintific	Rp 110.000	13	Sunscreen
24	AHA BHA PHA Toner	Skintific	Rp 105.000	14	Toner
25	5x Ceramide Barrier	Skintific	Rp 115.000	15	Moisturizer

Gambar 1.24 Output Ini Fitur Ke 2 Pada Menu Melihat Produk Yaitu Menampilkan Produk Dari Z -> A (Menu Admin)

[Harga Termurah]

No	Nama Produk	Merk	Harga	Stok	Jenis
1	Bright Stuff Face Wash	Emina	Rp 28.000	40	Face Wash
2	Face Wash Aloe	Wardah	Rp 30.000	25	Face Wash
3	Gentle Face Wash	Glad2Glow	Rp 35.000	20	Face Wash
4	Bright Stuff Toner	Emina	Rp 35.000	45	Toner
5	Ms Pimple Moisturizer	Emina	Rp 36.000	30	Moisturizer
6	Hydrating Toner	Wardah	Rp 38.000	30	Toner
7	Sun Protection SPF30	Emina	Rp 40.000	50	Sunscreen
8	Acne Care Serum	Emina	Rp 42.000	28	Serum
9	Moisturizer Gel	Glad2Glow	Rp 45.000	30	Moisturizer
10	UV Shield Sunscreen	Wardah	Rp 48.000	32	Sunscreen
11	Daily Moisturizer	Wardah	Rp 52.000	20	Moisturizer
12	Daily Sun Protect SPF35	Glad2Glow	Rp 53.000	22	Sunscreen
13	Lightening Serum	Wardah	Rp 55.000	40	Serum
14	Exfo Toner AHA BHA	Glad2Glow	Rp 57.000	20	Toner
15	Mild Cleanser FaceWash	Originote	Rp 60.000	15	Face Wash
16	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
17	Hyal Toner Essence	Originote	Rp 65.000	18	Toner
18	Niacinamide Serum	Originote	Rp 72.000	20	Serum
19	Hyalucera Moisturizer	Originote	Rp 75.000	22	Moisturizer
20	UV Barrier Sunscreen	Originote	Rp 78.000	16	Sunscreen
21	Acne Facial Wash	Skintific	Rp 95.000	14	Face Wash
22	AHA BHA PHA Toner	Skintific	Rp 105.000	14	Toner
23	AMPM Sunscreen SPF50	Skintific	Rp 110.000	13	Sunscreen
24	5x Ceramide Barrier	Skintific	Rp 115.000	15	Moisturizer
25	SymWhite Serum	Skintific	Rp 125.000	12	Serum

Gambar 1.25 Output Tampilan Fitur Yang Ke 3 Pada Menu Lihat Produk Yaitu Menampilkan Produk diStore kami Dari Harga yang Termurah -> Termahal (Menu Admin)

[Stok Sedikit]

No	Nama Produk	Merk	Harga	Stok	Jenis
1	SymWhite Serum	Skintific	Rp 125.000	12	Serum
2	AMPM Sunscreen SPF50	Skintific	Rp 110.000	13	Sunscreen
3	AHA BHA PHA Toner	Skintific	Rp 105.000	14	Toner
4	Acne Facial Wash	Skintific	Rp 95.000	14	Face Wash
5	Mild Cleanser FaceWash	Originote	Rp 60.000	15	Face Wash
6	5x Ceramide Barrier	Skintific	Rp 115.000	15	Moisturizer
7	UV Barrier Sunscreen	Originote	Rp 78.000	16	Sunscreen
8	Hyal Toner Essence	Originote	Rp 65.000	18	Toner
9	Daily Moisturizer	Wardah	Rp 52.000	20	Moisturizer
10	Exfo Toner AHA BHA	Glad2Glow	Rp 57.000	20	Toner
11	Gentle Face Wash	Glad2Glow	Rp 35.000	20	Face Wash
12	Niacinamide Serum	Originote	Rp 72.000	20	Serum
13	Daily Sun Protect SPF35	Glad2Glow	Rp 53.000	22	Sunscreen
14	Hyalucera Moisturizer	Originote	Rp 75.000	22	Moisturizer
15	Face Wash Aloe	Wardah	Rp 30.000	25	Face Wash
16	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
17	Acne Care Serum	Emina	Rp 42.000	28	Serum
18	Hydrating Toner	Wardah	Rp 38.000	30	Toner
19	Moisturizer Gel	Glad2Glow	Rp 45.000	30	Moisturizer
20	Ms Pimple Moisturizer	Emina	Rp 36.000	30	Moisturizer
21	UV Shield Sunscreen	Wardah	Rp 48.000	32	Sunscreen
22	Lightening Serum	Wardah	Rp 55.000	40	Serum
23	Bright Stuff Face Wash	Emina	Rp 28.000	40	Face Wash
24	Bright Stuff Toner	Emina	Rp 35.000	45	Toner
25	Sun Protection SPF30	Emina	Rp 40.000	50	Sunscreen

Gambar 1.26 Output Tampilan Fitur Yang ke 4 Pada MenU Lihat Produk Yaitu Untuk Mellihat Produk di Store Kami Dari Yang Paling Sedikit Hingga Yang Paling Banyak (Menu Admin)

CARI PRODUK BY JENIS

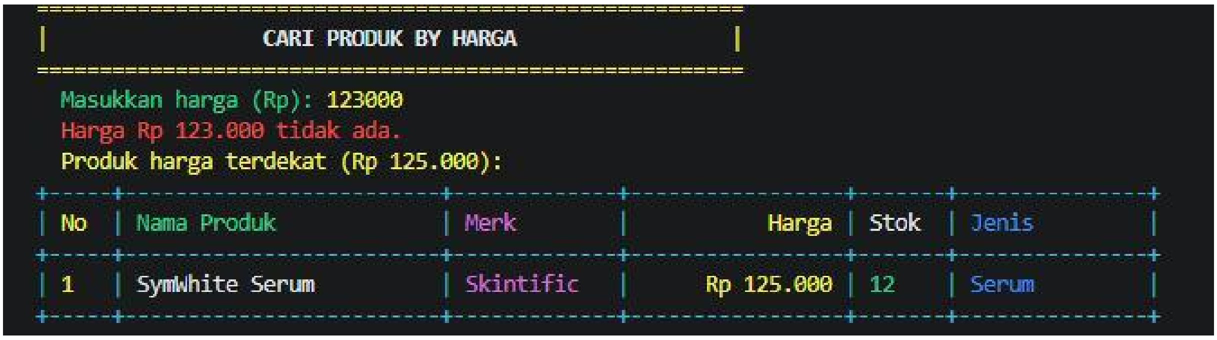
Jenis tersedia : Serum, Toner, Moisturizer, Face Wash, Sunscreen
(ketik semua untuk tampilkan semua jenis)

Cari jenis skincare: serum

No	Nama Produk	Merk	Harga	Stok	Jenis
1	Lightening Serum	Wardah	Rp 55.000	40	Serum
2	Brightening Serum	Glad2Glow	Rp 62.000	25	Serum
3	Acne Care Serum	Emina	Rp 42.000	28	Serum
4	Niacinamide Serum	Originote	Rp 72.000	20	Serum
5	SymWhite Serum	Skintific	Rp 125.000	12	Serum

[OK] Ditemukan 5 produk untuk jenis "serum".

Gambar 1.27 Output Fitur ke 5 Pada Menu Lihat Produk Dengan Mencari By Jenis (Menu Admin)

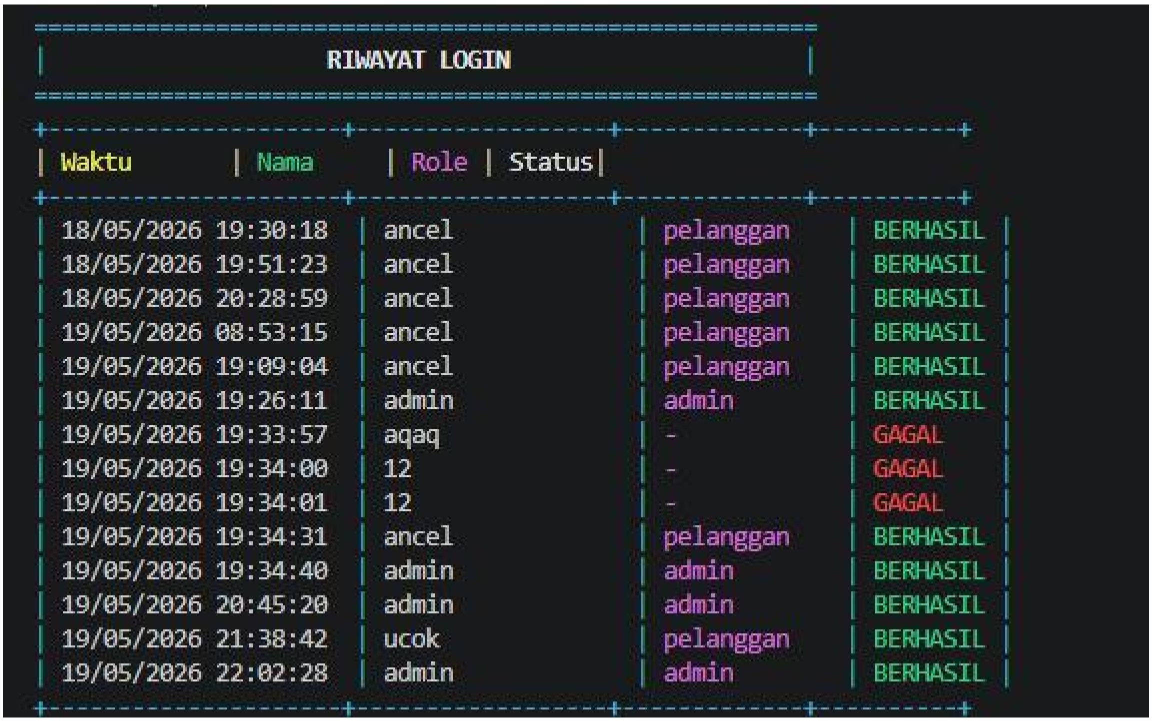


CARI PRODUK BY HARGA

Masukkan harga (Rp): 123000
Harga Rp 123.000 tidak ada.
Produk harga terdekat (Rp 125.000):

No	Nama Produk	Merk	Harga	Stok	Jenis
1	SymWhite Serum	Skintific	Rp 125.000	12	Serum

Gambar 1.26 Output Fitur Ke 6 Pada Menu Lihat Produk fitur Ini Untuk Mencari Produk By Harga (Menu Admin)



RIWAYAT LOGIN

Waktu	Nama	Role	Status
18/05/2026 19:30:18	ancel	pelanggan	BERHASIL
18/05/2026 19:51:23	ancel	pelanggan	BERHASIL
18/05/2026 20:28:59	ancel	pelanggan	BERHASIL
19/05/2026 08:53:15	ancel	pelanggan	BERHASIL
19/05/2026 19:09:04	ancel	pelanggan	BERHASIL
19/05/2026 19:26:11	admin	admin	BERHASIL
19/05/2026 19:33:57	aqaq	-	GAGAL
19/05/2026 19:34:00	12	-	GAGAL
19/05/2026 19:34:01	12	-	GAGAL
19/05/2026 19:34:31	ancel	pelanggan	BERHASIL
19/05/2026 19:34:40	admin	admin	BERHASIL
19/05/2026 20:45:20	admin	admin	BERHASIL
19/05/2026 21:38:42	ucok	pelanggan	BERHASIL
19/05/2026 22:02:28	admin	admin	BERHASIL

Gambar 1.27 Output Tampilan Menu Riwayat Login Pada Menu Admin

5. Langkah - Langkah GIT

5.1 GIT INIT

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD> git init
Reinitialized existing Git repository in C:/Users/lenovoindo/OneDrive/Documents/Praktikum-APD/.git/
```

Untuk membuat repositori Git baru di sebuah direktori proyek, baik itu proyek yang baru atau proyek yang sudah ada dan belum dikelola oleh Git.

5.2 GIT ADD

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD> git add .
```

perintah di Git untuk menyiapkan atau "staging" perubahan file Anda untuk dimasukkan ke dalam riwayat proyek (commit berikutnya).

5.3 GIT COMMIT

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git commit -m "post test 3"
On branch main
nothing to commit, working tree clean
```

Untuk menyimpan (merekam) perubahan yang telah Anda buat pada file ke dalam riwayat proyek Anda, menciptakan "snapshot" dari proyek pada titik waktu tertentu.

5.4 GIT REMOTE

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git remote add origin https://github.com/anza-cyber/Praktikum
-APD-c1.git
```

Untuk berbagi perubahan kode, mengelola versi proyek, dan berkolaborasi secara efektif melalui operasi push dan pull.

5.5 GIT PUSH

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.00 KiB | 293.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/anza-cyber/Praktikum-APD-c1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3>
```

Untuk mengunggah perubahan dari repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga perubahan tersebut dapat dibagikan dengan orang lain atau disinkronkan dengan versi proyek yang lebih besar.

6. Pembagian Tugas

- Rafa :

1. Menu Pelanggan
2. Error Handling
3. Tampilan Output
4. Memperbaiki Semua Fitur
5. Flowchart Menu Admin
6. Laporan

- Ancel

1. Riwayat Login
2. Riwayat Data Pengguna
3. Struk Pembelian
4. Konfirmasi y/n
5. Flowchart Menu Admin
6. Metode Pembayaran

- Dava

1. CRUD Menu Admin
2. Memperbaiki Fitur Program
3. Sorting
4. Searching
5. Membuat Konsep Program
6. Flowchart Menu Pelanggan